Hydrometric Monitoring

Restoration Monitoring Guidance

12 September 2025

Document details	This document provides general guidance on hydrometric monitoring, primarily for the purpose of assessing the effects of restoration activities
Document title	Hydrometric Monitoring
Document subtitle	Restoration Monitoring Guidance
Date	12 September 2025
Version	0.3
Author	Nicci Zargarpour

Document history

Version	Revision	Author	Reviewed by	Date	Comments
Draft	0.1	Nicci Zargarpour	Oliver Franklin	06-June-24	
Draft	0.2	NZ		20-June-24	Comments addressed, changes tracked
Draft	0.3	NZ	Oliver Franklin	12-Sep-25	Previous comments accepted/cleared. New changes are tracked. Updated based on comments from Paula Soto and training with the Province.

CONTENTS

Acronyms and Abbreviations

ADV	Acoustic Doppler Velocity
BDA	Beaver Dam Analogue
BM	Benchmark
BS	Backsight
ECCC	Environment and Climate Change Canada
FS	Foresight
LSPIV	Large-Scale Particle Image Velocimetry
PFD	Personal Flotation Device
RISC	Resources Information Standards Committee
TP	Turning point
USEPA	U.S. Environmental Protection Agency
QAQC	Quality Assurance and Quality Control
WSC	Water Survey of Canada

1. OVERVIEW

The purpose of this document is to provide guidance on how to collect accurate, continuous hydrologic data at wadeable stream sites in support of fish habitat restoration monitoring projects. This guidance facilitates comparisons of continuous hydrologic data among sites and ensures data are of sufficient quality for future analyses. Much of the content within this document is sourced from the Resources Information Standards Committee (RISC 2018) Manual of British Columbia Hydrometric Standards, Version 2.0 which forms the benchmark for hydrometric monitoring in British Columbia and covers a broad range of methodologies and technologies. This guidance document has been tailored to the contexts and techniques that are often associated with fish habitat restoration monitoring, and is structured as follows:

- **Section 1** includes restoration-specific perspectives and key definitions.
- **Section 2** describes how to select an appropriate site for establishing a hydrometric monitoring station.
- **Section 3** describes how to install and maintain pressure transducers and staff gauges to measure continuous stage (water level).
- **Section 4** describes how to collect the discharge measurements that will be used to develop stage-discharge curves.
- **Section 5** briefly discusses rating curve development.
- **Appendix A** provides a stand-alone, condensed, step-by-step guide that outlines activities that field technicians should complete prior to and during a typical site visit when establishing hydrometric monitoring stations and when completing repeat visits.

1.1 Hydrometric Monitoring for Reach-Scale Restoration

Monitoring of restoration projects, which often occur at the scale of a reach rather than the watershed, may require a distinct perspective from most hydrometric study designs. Hydrometric stations are installed for a variety of purposes: monitoring baseline trends in flow, exploring watershed-scale causes of variation in discharge (land and water use changes, climatic fluctuations, etc.), predicting flood conditions, etc. Hydrometric stations are typically positioned to obtain flow data that is representative of a catchment area, and may be maintained to be free of 'local' disturbances such as logs that elevate the water level at the stage logger. In the case of reach-scale restoration, these local disturbances may actually be, or be a consequence of, our restoration actions, and their effects on discharge or stage may be the reason for installation of our hydrometric station.

If we are interested in how water behaves within our restored reach, rather than the entire watershed, the type and location of data collection may differ. In some cases, for restoration monitoring purposes it may be sufficient to obtain only stage data. However, where possible, it will be of value to also obtain discharge data. Discharge data can inform project design and generate widely-used statistics relating to flow (e.g. critical environmental flow needs thresholds, flood levels, water use impacts), and may also be useful for liability concerns (e.g., demonstrating to property owners that projected flood levels did not increase).

Restoration project planning and adaptive management benefits from knowledge of water elevation and the anticipated range of discharge conditions. This information is estimated using a stage-discharge rating curve, and can inform decisions regarding what structures, features, or sediment sizes will persist under expected flow conditions. It is generally recommended to develop a stage-discharge rating curve to convert continuous water level data into flow data, as the resources committed to obtain discharge measurements at the project site under a range of flow conditions may be insignificant compared with replacement of structures, substrates, or unplanned flooding. Similarly, the properties of the stage-discharge rating curve itself can provide helpful information on how flow and habitat may be related.

Recognising that it is not always feasible to develop stage-discharge rating curves, here we provide guidance specific to restoration monitoring, including some general examples of cases where stage, discharge, and stage-discharge rating curves will be beneficial. In some cases, it may be of interest to combine multiple approaches: For example, we may install stage-loggers to monitor water level in an area of restored stream geometry, and also establish a stage-discharge rating curve at a location upstream of the restored geometry. This combined approach may be necessary if we are interested in how water levels respond to discharge in an area that is not conducive to obtaining discharge measurements (e.g., anastomosing channels, see Section 4.3 for Site Selection considerations).

1.1.1 Stage-Discharge for Reach-Scale Restoration

If our restoration objectives involve intentional (rather than incidental) modifications of the flow regime, with effects transmitted downstream of our restored reach, stage-discharge rating curves (and the resulting discharge series they can help produce) will be more informative than stage alone. For example, for Beaver Dam Analogue (BDA) complexes or similar floodplain engagement approaches, it is recommended to develop stage-discharge curves upstream and downstream of the restored reach. This way we can quantify the extent to which the BDA or floodplain complex may increase discharge during dry periods compared with its upstream input (i.e., acting like a sponge). Without the stage-discharge curve and resulting discharge series, we are limited to stage information, which permits only qualitative inferences regarding discharge.

1.1.2 Stage Data for Reach-Scale Restoration

If our restoration objectives seek to alter flow conditions or water level within our restoration reach without altering flows downstream, it may be sufficient to obtain stage data without creating a stage-discharge rating curve. Note that a current meter might still be used to collect informative flow velocity data at particular focal locations, but this approach is without the requirement for multiple return visits under a variety of flow conditions. Some examples of monitoring using stage loggers without a rating curve include:

- Comparisons of water level at focal locations (e.g., rearing or holding pools) with water level from before restoration, from control/reference sites, from a main channel, or compared to benchmarks anticipated to determine accessible or suitable habitat. Metrics may include water depths within a range of reference conditions, a directional change in depth from a degraded control site, duration and timing of water above a given depth (or duration and timing that habitat is wetted/inundated) etc. This approach may be particularly useful in inferring the presence and duration of flow where channels or off-channel habitat are restored/created, or to evaluate flow-diversion structure performance.
    - Note that stage loggers will not distinguish between flowing and stagnating water. If we are interested in identifying the timing/duration of *flowing* water, we can measure the residual depth (maximum depth – riffle crest depth) of the pool in which the logger is deployed, and interpret stage levels below this depth as non-flowing. Note that this would be a conservative estimate of flow cessation, as flow between gravels may persist.
- Indirect qualitative inferences regarding the rate of flow through a restored reach are possible. For example, comparisons of stage upstream and downstream of a reach from before to after restoration: The time it takes a peak in flow to travel from the upstream logger to the downstream logger may be altered by restoration efforts. If this effect is consistently observed, it may demonstrate that the restoration effort has changed the rate of flow. However, such changes would likely only be detectable under quite extreme changes to the stream and requires that stage is recorded more frequently than typical.
- Where available, direct measurements of water velocity can be associated with a particular stage (without providing the ability to extrapolate to different, unobserved discharge conditions). This

may be appropriate for demonstrating the suitability of flow conditions for fish, particularly if there is a potential bottleneck or a particular water level at which fish passage is anticipated.

- Combining topographic information with stage can permit projections of inundation based on recorded water levels. This may address an objective relating to floodplain reconnection extent and/or duration. Beyond this, one might obtain targeted velocity measurements to evaluate the suitability of flow conditions in inundated habitats and in the main channel, and compare these ‘point values’ to data from before the floodplain reconnection.

1.1.3 Discharge Data for Reach-Scale Restoration

Discharge series are estimates of discharge across time, derived from the stage-discharge rating curve and the recorded stage time series. Discharge series have a wide range of applications, which includes restoration monitoring at the reach-scale or beyond. For example, discharge series can be used to assess and interpret changes in other important habitat or fish data, such as determining the flow conditions at which water temperatures and dissolved oxygen levels are optimal for particular salmon life stages, identifying at which flows off-channel habitat becomes (dis)engaged, or determining at what flows or times of year barriers may become hydraulically impassable to fish.

The ability to relate discharge (rather than water level) to physical and biological observations can inform adaptive management and also provide context by connecting flow at the restoration site with broader flow information (e.g., providing reassurance that observed flows represent flood events of a given frequency, and that these flows were contained within the restored reach).

1.2 Key Terminology

This section briefly covers important concepts and definitions relevant to this protocol

Backsight (BS)

A sight taken on a benchmark or point of known elevation in order to determine the instrument height.

Backwater

Backwater is defined as a temporary rise in stage produced by an obstruction in the stream channel downstream of the gauge caused by ice, weeds, control structure, etc. The difference between the observed stage for a certain discharge and the stage as indicated by the stage-discharge relation for the same discharge is reported as the backwater at the station.

Benchmark (BM)

A permanent, fixed reference point of known elevation.

Control

The condition downstream from a gauging station that determines the stage-discharge relation. Natural controls might include a downstream riffle, bedrock outcrop, or other stream feature that controls flow. Unnatural controls might include a bridge, culvert, weir, or other artificial structure that is narrower than the stream channel and constricts flow. Note, the feature controlling the stage-discharge relationship can change at different flow levels. These changes will be reflected in the rating curve.

Closure Error

The difference between the elevation of the starting point of a closed level circuit and the elevation of that same point from the final instrument setup.

Collimation Error

The deviation of a level's line of sight from horizontal, often given as a vertical deviation per horizontal distance, such as x millimetres per y metres.

Discharge, Q

The volume of liquid flowing through a cross-section per unit of time.

Discharge measurement

The determination of the rate of discharge at a gauging station on a stream, including an observation of "no flow", which is classed as a discharge measurement.

Discharge rating (or Stage-discharge relation)

A curve (or rating equation/s) that expresses the relation between the stage and the discharge in an open channel at a given stream cross-section.

Foresight (FS)

A sight taken on a point for which an elevation is to be determined.

Gauge Correction

A correction that is applied to recorded water levels to account for vertical movement of the reference gauge.

Gauge Datum

The surface to which gauge height elevations are referenced. It is also referred to as the operating datum.

Gauge Height

Gauge height is the height of the water surface above the gauge datum. Gauge height will be collected during every site visit and may be obtained a variety of ways:

- Levelling surveys from bench marks
- Direct visual observation of a staff gauge

This term is used interchangeably with stage and water level.

Height of Instrument

The elevation of the horizontal line of sight of a surveying instrument.

Hydrometric Monitoring Station

A hydrometric monitoring station (also referred to as a gauging station or a hydrometric station) is a natural or constructed location on a watercourse where records of stage (water level) and discharge (stream flow) are systematically obtained.

Level Circuit

A line of levels that ends at the point of origin or at another previously established benchmark.

Levelling rod

A graduated rod used in measuring the vertical distance between the point of interest (e.g. Benchmark or water level) and the height of instrument.

Staff gauge

A manual gauge consisting of a graduated plate or rod that is set vertically in the streambed or attached to a solid structure.

Stage

The elevation of the free surface of a stream, lake, or reservoir relative to a gauge datum. It is used interchangeably with the terms “gauge height” and “water level”.

Stilling well

A well (tube) connected with the stream in such a way as to permit the measurement of the stage in relatively still conditions (natural surging is dampened).

Turning point (TP)

A fixed point on which a foresight rod reading is taken, then the instrument is moved, and a backsight rod reading is taken in order to establish a new instrument height.

Velocity-area method

Method of discharge determination deduced from the area of the cross-section, bounded by the wetted perimeter and the free surface, and the integration of the component velocities in the cross-section.

2. STATION SELECTION AND INSTALLATION

A hydrometric monitoring station is a natural or constructed location on a watercourse where records of stage and discharge are systematically obtained.

A well-planned and constructed hydrometric monitoring station meets the following criteria:

1. It is possible to get an accurate water level reading from the gauge at all stages.
2. The control, whether natural or artificial, is stable.
    a. A stable natural control may be a:
        i. bedrock outcrop, or other stable riffle (shoal) for measuring during low flow
        ii. channel constriction for measuring at high flow
        iii. falls or cascade that is not submerged at any water level
        iv. long reach of stable bed
    b. A stable artificial control may be a:
        i. rated structure (flume, weir, etc.)
        ii. fish barrier
        iii. streambed sill (log, concrete, etc.)
3. Discharge can be measured accurately at all stages, either with a current meter or by other means.

4. The site access is secure and safe during the operational season.
5. The station is structurally sound (e.g., can withstand being overtopped).

2.2 Desktop Reconnaissance

To select an appropriate site for establishing a hydrometric monitoring station, it is critical to identify the purpose of the station.

The watercourse and basin should be studied prior to initiating any field trips. Resources and techniques that may be used to gather relevant information include:

- Reviewing topographical and geological maps.
- Reviewing available high resolution aerial imagery.
- Seeking out individuals that are familiar with the watercourse and the region to discuss flood history and activities in the area that might affect the watercourse (e.g., logging, reservoirs, bridge construction, diversions etc.).
- Downloading and reviewing relevant data available from regional Water Survey of Canada (WSC) hydrometric stations.

Ideally, sites for a hydrometric monitoring station should have the following characteristics:

- Location of a suitable control, either natural or artificial
- Stable channel, no likelihood of degradations and aggregation and free from aquatic vegetation
- Straight, aligned stream banks that contain all flows
- If performing discharge measurements to develop a stage-discharge curve, the current meter measuring site should be within reasonable proximity to the gauge. Good current meter measuring sites typically feature flow confined to one channel at all stages, no undercut banks, minimal obstructions, no turbulence, no slow-moving pools (dead water or backwater effects), no eddies.
- Site access:
    - Safe access to the site over the full range of measurements
    - Ability to carry materials to the site
    - A long-term agreement with any land owner whose land must be crossed to gain access to the site
- No inflows between gauge and metering sites
- No wetlands or pooling of water downstream or in the vicinity of gauge.

Look for the characteristics of a suitable site and select two or three candidate locations. Prepare a plan to visit the site identified as having the most potential, though be prepared to move on to other potential sites if the first one does not work out.

2.3 Field Reconnaissance

Ideally, selection of a gauging station is made with a field reconnaissance trip to the potential sites. However, it is recognized that this may not always be possible, and site selection may need to occur during the site installation visit. The field reconnaissance visit is an opportunity to evaluate the potential site(s) based on the objectives and characteristics described in Section 2.2. This site visit should include careful observation of the following:

- Low Flow Conditions:

- Look for a stable, well-defined low water control. If a stable control is not available, consider the feasibility of building an artificial low water control. Investigate the options, and gather and record preliminary survey data.
- If a site with a movable streambed must be accepted (i.e. one with a mobile granular channel), it is best to locate the gauge in as uniform a reach as possible, avoiding any channel obstruction that may intensify scour and fill.
- Where the watercourse emerges from an area of steeper gradients onto an alluvial fan, the reconnaissance must include discharge measurements to determine where the seepage of water into the alluvium becomes significant. The station should be located upstream of the area of water seepage if the maximum yield of the watercourse is required.
- Select a location that allows for the stage datalogger to remain in sufficient water depth during periods of low flow.
- If your location of interest includes monitoring of areas of low flow or potential flow disconnection, for example where fish passage may be a concern, it may be advisable to supplement your flow data with trailcam-derived stream connectivity data (e.g., Bellucci *et al.* 2020, Zargarpour and Franklin 2025). This can provide quasi-quantitative data of surface connectivity and fish passage inferences at periods where water levels are too low to be detectable with a stage datalogger.

■ High Flow Conditions:

- Look for evidence of past flooding (e.g., trash lines) to determine the magnitude of maximum discharge. Consider if the site could be safely accessed at all stages.

■ Flow Measurement Conditions:

- If performing discharge measurements to develop a stage-discharge rating curve, ensure that the site has a suitable location for measuring discharge, either with a current meter or by other means (Section 4.3).

2.4 Preparation for Station Installation

The field reconnaissance visit will help to inform the preparation of the preliminary conceptual designs and costs. However, with or without a reconnaissance visit, the factors to consider when producing a cost estimate are:

■ Anticipated period of recording and/or the period of operation (seasonal, low flow, etc.)

■ Requirements for an artificial control in the absence of channel stability

- The availability of suitable native material at or near the site
- Construction methods

■ The cost of transporting personnel, equipment, and materials to the site

■ Installation methods

■ Selected instrumentation

■ Scheduling constraints based on flow magnitude and fishery regulations for in-stream works.

■ Future operational resources

2.5 Station Installation

2.5.1 Establishment of Gauge Datum

The gauge datum is a conceptual “surface” or “plane”, and the zero (0.000 m) from which all measurements of stage are referenced. For convenience, the gauge datum should be below the lowest water for lakes or below zero flow for streams, so that all stage readings are positive numbers. Following station installation, all hydrometric data will be adjusted to height of the water surface above gauge datum, called stage or gauge height.

The continuity of the gauge datum at a gauging station is an important part of data collection, and every effort should be made to maintain it throughout the entire period of operation. If datum changes are made for whatever reason, the change should be well documented. It is critical that the gauge datum is established as soon as a station is installed. This is done through the installation of a network of three or more stable bench marks. The datum at each gauging station should be periodically checked by running levels from the benchmarks.

2.5.2 Benchmarks

Benchmarks are permanent, fixed reference points of known elevation installed when gauging stations are first established. Benchmarks are used for:

- Establishing the gauge datum
- Maintaining the datum to ensure continuity of the record
- Facilitating the confirmation of gauge height or stage relative to a constant datum or reference elevation. Gauge heights are used for adjusting recorded stage data (often recorded as water depth above the sensor) to water surface elevation above the gauge datum

The main requirement for benchmarks used in hydrometric operations is that they are stable, readily identifiable, and accessible at all water levels. Types of benchmarks include:

- Lag bolt in bedrock or fixed structures
- Lag bolt in a tree
- Brass tablet or lead plug attached vertically or horizontally in a rock or concrete surface

When installing benchmarks it is critical to select sites that will provide a high degree of stability (i.e., maintain its relative position in the local terrain). Benchmarks must be spread out, away from the river bank, and preferably above the maximum expected water level. Once established, benchmarks must each have a unique identifier (ID), and be clearly marked and documented so that they will be easy to find at a later date. A minimum of three independent benchmarks is required at each gauging station for redundancy, and to enable confirmation of the stability of the benchmarks.

The initial benchmark levelling survey is used to establish gauge datum (Section 2.5.1). The general approach is as follows:

1. Select a primary benchmark (choose the most stable/permanent benchmark), this benchmark will be assigned an elevation of 100.000 m above the gauge datum.
2. Other benchmarks are then surveyed and assigned their value relative to the primary bench mark.

In some instances, temporary benchmarks may be required. Temporary benchmarks are used for only relatively short periods, usually until permanent benchmarks can be established or re-established. The purpose of the temporary benchmark is to ensure continuity of gauge datum when the temporary benchmarks are tied in to the permanent benchmarks. They are often used for newly constructed stations and for existing stations at which permanent benchmarks have recently been destroyed. Lag bolts and nails or marks on stable structures are commonly used to establish temporary benchmarks.

2.5.2.2 Benchmark Installation and Levelling

Levelling is the process by which elevations of points of interest are determined. Levelling for hydrometric operations serves four main purposes:

1. Monitoring stability of a system of benchmarks
2. Monitoring vertical displacement of a reference gauge to determine gauge corrections
3. Performing direct water level measurements to determine sensor reset corrections
4. Obtaining elevations of other points such as a high-water mark or the elevation of the riverbed.

All levelling activities rely on benchmark stability monitoring since the elevation of the primary benchmark is used to determine the elevations of all other surveyed points.

A benchmark or reference gauge is considered stable if:

- Its surveyed elevation has fluctuated about a consistent value throughout the last three years and the deviation from the mean surveyed elevation has not exceeded the survey tolerance.

A benchmark or reference gauge is considered unstable if:

- Its surveyed elevation differs from its established elevation by more than the survey tolerance, and/or there is a trend in the deviation of the surveyed elevation (even if the deviation is within the survey tolerance).

A station’s local reference network (i.e., the assemblage of benchmarks at a site) is classified as stable if all benchmarks in that network are stable. Having a stable local reference network will ensure that the continuity of the operating datum is maintained.

Unstable benchmarks or reference gauges require more frequent levelling and may result in the application of gauge corrections when used for water level measurements.

The steps outlined below describe the procedures and record-keeping used when installing benchmarks and establishing gauge datum. This section assumes that three clearly identifiable benchmarks have been installed.

Step 1 – Set up the levelling instruments

Levelling instruments consist of a survey level mounted on top of a tripod, and a telescopic stadia rod. Note, it is strongly recommended that levelling instruments are frequently checked for accuracy. A levelling test (i.e., Two-peg test) should be carried out prior to any site visits, and after any rough usage (e.g., transportation). Refer to Appendix B for detailed instructions.

- The tripod should be set on firm ground so that the technician operating the level has secure footing and can conduct the level circuit. The level and tripod should be placed in a location approximately midway between the backsight (BS; point of known elevation) and foresight (FS; point of unknown elevation), with the tripod legs pressed firmly in the ground and the height adjusted appropriately for sighting (Figure 1). Using the levelling screws on the base, adjust the level such that the circular bubble is centred within the setting circle.

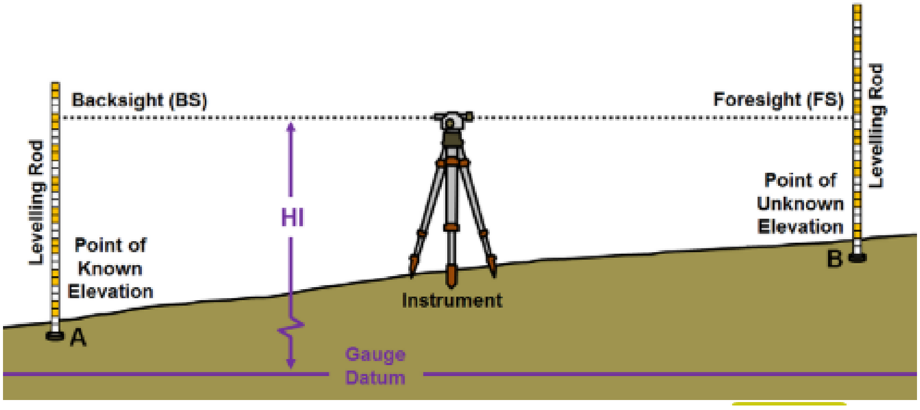

Figure 1: Illustration of Instrument Setup for Levelling (Source: RISC 2018)

Step 2 – Position and read the stadia rod

- Position the stadia rod vertically on the point of a stable object (e.g., benchmark). For horizontally mounted benchmarks, take care to position the rod on the highest point for the reading. The following techniques can be used to ensure the rod is plumb:
  - Use a rod level, or
  - If a rod level is not available, wave the rod slowly towards and away from the instrument and observe the lowest reading on the rod
- Observe and record the reading indicated by the horizontal crosshair on the rod as seen through the telescope.
- Record detailed field notes, sketches and data within the Hydrometric Levelling Survey Form (Appendix E). The Hydrometric level note forms are divided into columns for recording observations and calculating elevations.
  - The 'Station' column is used to record the name of the objective point (e.g., benchmark identification number, turning point number, type of reference gauge). Information recorded on each horizontal line pertains to that objective point.
  - There are columns for 'Backsight', 'Height of Instrument', 'Foresight', 'Elevation' and 'Notes'.

Refer to Appendix A, Section A.2.6 for a detailed example of the documentation process for a levelling survey.

2.5.2.3 Levelling Errors

When a level circuit is closed, the surveyed elevation of the last point (which is also the first point) may not equal the elevation used at the start of the circuit due to sources of error. This difference in elevation is called the closure error. The maximum allowable closure error is referred to as the survey tolerance. For levelling surveys with three or fewer setups, the survey tolerance is ± 0.003 m. If the absolute value of the closure error > 0.003 m, the circuit should be rejected and redone until a closure error of ≤ 0.003 m is achieved.

The following section describes common errors in levelling and the steps field technicians can take to limit them:

- **Improper adjustment of an instrument:** This condition occurs when the line of sight is not parallel to the axis of the level tube. This error can be minimized by careful adjustment of the instrument and by balancing backsight and foresight distances. Performing the two-peg test routinely will help detect this error (Appendix B).

- **Parallax:** This condition occurs when there is an apparent movement of the crosshairs on the target with a corresponding slight movement of the observer's eye (vertically or horizontally). This condition can be minimized by carefully focusing the eyepiece on the objective lens. Alternatively, focus the telescope to the greatest distance, tilt the level to view the sky, and adjust the eyepiece until the crosshairs are as sharp as possible.
- **Inaccurate reading of the levelling rod**: This error can be reduced by using shorter sights and by checking each reading before recording it.
- **Levelling rod is not plumb:** A rod level can be used or the rod can be waved towards and away from the instrument to minimize this type of error.
- **Improper turning points:** turning points that are not well defined and/or not stable are potential sources of error. Care should be taken to select appropriate turning points.
- **Settlement of the instrument tripod:** settlement of the tripod may occur when levelling over soft, muddy, or thawing ground. Under these conditions, backsight and foresight observations should be made in quick succession to minimize any effect from the instrument settling.
- **Incorrect levelling rod length:** Dirt or snow can become trapped in the sleeve of sectional rods which may prevent sections from extending properly. Level rods should be periodically checked with a measuring tape and they should be carried and assembled so that the upper portion is never in contact with the ground. Mud, snow or ice accumulations on the rod must be removed before each reading is taken.

2.5.3 Reference Gauge Installation

All gauging stations require some form of water level reference gauge in addition to any water level recording device that may also be installed. Readings from the stage sensor are routinely compared to the readings from the reference gauge at each station. The sensor reading must be referenced to this gauge, which in turn is referenced to station datum with benchmarks. The most common reference gauge is the vertical staff gauge (Section 2.5.3.1). However, a benchmark (Section 2.5.2) can also be a reference gauge when stage is observed by direct levelling from it to the water surface; it is referred to as a direct water level. If a direct water level is used as the reference water level measurement, it should be taken at the same location on each site visit (note the measurement location in the station history).

2.5.3.1 Staff Gauge

The most commonly used reference gauge is the vertical staff gauge, which consists of one or more 1 metre sections of enamelled steel plate accurately graduated to either 0.01 or 0.002 m increments. Each decimetre is numbered and observers can estimate the third decimal value by reading the bottom of the meniscus. Practitioners may sometimes perceive the staff gauge reading as a simplistic task, spending little time taking the reading and subsequently misreading the gauge. For this reason, photographs or video of the staff gauge should be taken during every site visit so that the accuracy and precision of the reading can be verified at a later date, if necessary.

Staff gauges should be installed such that they are protected from damage by moving ice or other floating debris and are not affected by a local drawdown or pileup of water. Ideally, the gauge should be mounted so that the face is parallel to the current to reduce small local effects. To reduce velocity head difference on the up and downstream sides of the gauge, a length of half round wood moulding can be attached to streamline the upstream edge of the backing board.

When the gauge is to be used as a reference for a recording device, it should be positioned as close as possible to the sensor and in a zone of minimal local turbulence. Staff gauges should be installed carefully in order to ensure data continuity and ease of monitoring. Generally this involves:

1. Fastening the gauge plate(s) to a 0.15 m wide backing board.

2. Securing the backing board to a convenient vertical face such as a bridge abutment, pier, or piling. Note, attachments on concrete, bedrock, or steel structures require fastening tools, anchors, framing, or plumber's strapping. If no suitable anchor point is available, a steel angle iron driven into the streambed is acceptable. Slotted mounting holes on the gauge plate allow for limited vertical adjustment of the plate during the initial installation, and can be used to eliminate minor gauge corrections identified at subsequent gauge level checks.

2.5.3.2 Reference Gauge Checks

Reference gauges can be subject to extreme conditions and can become displaced or destroyed over time. To help ensure that stage records remain reliable, vertical movement of the reference gauge must be regularly monitored by running levels from the benchmarks (Section 2.5.2). Levels should be run between all benchmarks and reference gauges once a year at stable sites (ideally after spring thaw), and two times per year at more unstable sites. However, the elevation of a reference gauge with a history of instability may have to be monitored more frequently. When running levels to check for the possible movement of a reference gauge, the circuit must be closed and closure error must be within survey tolerance. Any changes should be documented in the station history.

Gauge Corrections

Gauge corrections compensate for temporary vertical movement of the reference gauge. They are computed each time that the reference gauge is surveyed by subtracting the established elevation (E_e) of the reference gauge from its surveyed elevation (E_s):

$$GC = E_s - E_e$$

If the absolute value of GC is ≤ 0.003 m, no correction should be applied. If it exceeds 0.003 m, a gauge correction is applied to all stage data. Gauge corrections are applied until the elevation of the reference gauge returns to its previously established elevation, until a new gauge correction is determined via levelling, or until a new established elevation is determined via Benchmark History Analysis. Detailed procedures for surveying the reference gauge elevation and determining gauge correction are described in the "Hydrometric Field Manual, Levelling" published by Environment and Climate Change Canada (ECCC) 2017.

3. STAGE MEASUREMENT

Measurement and recording of stage (water level) provides information about streamflow patterns, including the timing, duration, and frequency of high or low flows, and provides the means by which discharge computations are made. Stage data from hydrometric stations consists of both discrete, manual measurements, and continuously recorded measurements. Both measurement types are integral to the collection of continuous hydrologic data, with discrete manual measurements of accurately surveyed stage (gauge height) used to validate and correct continuously recorded stage data from installed instrumentation.

Instrumentation used to collect water level or stage height includes:

- manual gauges, and
- recording gauge or automated stage sensors for continuous monitoring of water levels.

3.1 Manual Gauges

In a manual gauging station, stage readings can be obtained from a permanently installed manual gauge (e.g., staff gauge, Section 2.5.3.1) or by the direct survey of water level referenced to the station's datum or benchmarks.

3.2 Continuous Recorders and Associated Instrumentation

Automated stage sensors are installed at gauging stations to automatically record the continuous stage. Ideally, all gauging stations should be equipped with automatic stage sensors. All continuous systems require frequent maintenance to ensure quality data collection, and manual reference to gauge heights during every site visit. There are a number of different methods for automated continuous measurement of stage. For the purpose of this guidance document, the sections below describe only the use of submersible pressure sensors (pressure transducers), as these types of sensors are commonly used and relatively simple to install. For information regarding other sensor types (including dry gas pressure sensors and ultrasonic and radar level sensors) refer to RISC (2018).

Submersible Pressure Sensors (Pressure Transducers) record absolute pressure (air pressure + water pressure), from which air pressure must be subtracted to yield water pressure. Software programs can then be used to convert water pressure to water level using the density of water.

Given atmospheric pressure changes with weather and altitude, these sensors require correction for barometric pressure. This can be accomplished in two ways:

1. If the transducer is vented, it has a vented cable that references and automatically corrects for atmospheric pressure; or,
2. If the transducer is non-vented, barometric pressure readings must be obtained from a separate device that is located nearby on land, and after both sets of data are downloaded, software is used to correct the water level data for the barometric variations.

Appendix C provides an overview of vented and non-vented pressure transducers typically used for hydrometric monitoring operations. Each system has advantages depending on the site conditions.

3.2.2 Selection of Sensors and Data Loggers

With the wide variety of data loggers available on the market, it is important for field technicians to consider which model will meet the site/project requirements. Some factors to consider are listed below:

- Programming capabilities (to acquire data at pre-set intervals)
    - Readings are typically at 15 minute intervals, starting at the top of the hour, and should not be less frequent. In general, a 15-minute interval will provide an accurate representation of actual stage behaviour for most streams.
- Ease of use
- Data transmission options
- Power supply requirements (i.e., internal power or solar recharge required)
- Manufacturer support
- Accuracy and resolution
- Ability to interface with the sensors
- Data processing capability (i.e., instantaneous, maximum-minimum, mean computations)
- Operational temperature range
- Operating system (i.e., Windows, Mac OS, Android)
- Telemetry capability (i.e., landline or cellular modem, radio, satellite, GOES)
- Reliability, compactness, cost
- Storage capacity

Hydrometric stations are often located in remote areas, and in some instances timely acquisition of data may be a high priority. In these situations, telemetry systems can be installed so that logged data are periodically transmitted to a receiver. Figure 2 illustrates the components that make up a typical electronic monitoring station system.

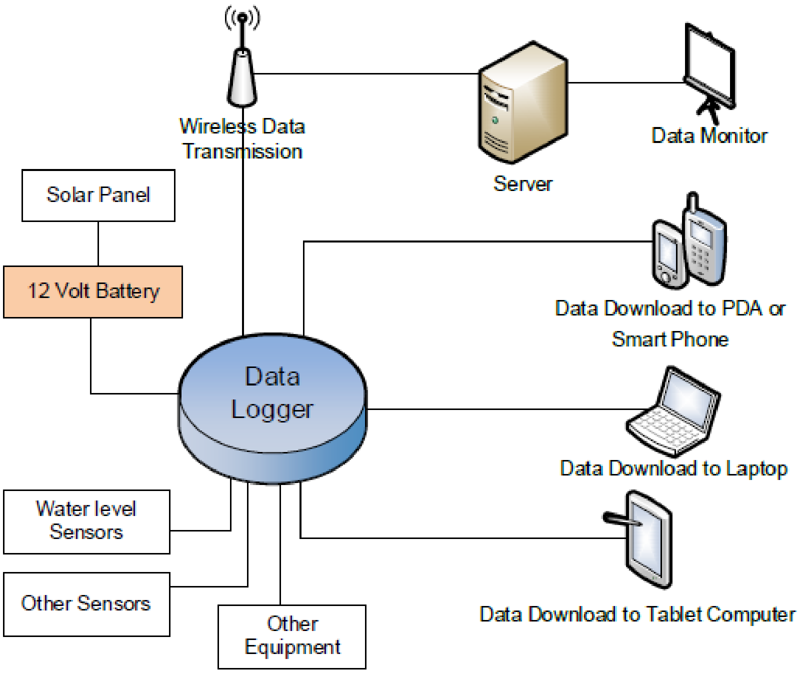

Figure 2: Typical flow diagram for electronic monitoring stations illustrating the components required to measure and log stage data (Source: RISC 2018)

3.2.3 Power Supply Considerations

Electronic stream monitoring equipment generally require a 12 V power supply for stations with multiple sensors. Commonly used power supply options include:

- Use of an existing on-site 110 AC power supply with a 12 V converter. Consider whether a backup power supply should be maintained at the site to support the station during power failures.
- Operate the site with battery(s) only. If the site is low-power, a heavy 12 V RV type or deep-cycle battery will likely operate the site for several months between charges.
- Use of solar panels and storage battery(s) may be necessary at sites where more power intensive data transmission components are required. When using solar power consider the following:
  - Determine the size of the panel required based on the total power requirements of all system electronics.
  - On-site, ensure the panel will have an unobstructed view of the sun through all months that the equipment will be operational.
  - For the system battery, use only heavy-duty deep-cycle 12V storage batteries.
  - Use a high-quality voltage regulator that also monitors the temperature of the storage battery

3.2.4 Documentation for Stage Measurements

Thorough documentation of the inspection record for water stage is extremely important as it establishes the identity of the location and the base values for future evaluation, and it ensures accurate transfer of the information from the field to the office. An inspection record for water stage should contain, at minimum:

- Station name and station identifier
- Date and time of the logger reference time
- Reference gauge readings
- A concurrent instantaneous value of logger or sensor

Refer to Appendix E for field forms.

3.2.5 Data Collection, Retrieval and Processing

Collection of stage data will depend on the type of recorder used and the data retrieval process of those systems. Telemetry systems (e.g., satellite, radio, cell-phone) can be used to transmit, decode, and enter data into a database. Alternatively, data can also be manually downloaded to a computer or mobile device during site visits. All stage data retrieved by manual download or transmitted via telemetry should be considered original data and must be preserved. Any corrections applied to raw data (due to site conditions or instrument malfunction etc.) should be recorded and archived with the data. The following information should be retained indefinitely:

- Unedited original raw data
- Final checked and verified data
- Associated metadata, including:
  - Data comments
  - Site details
  - Recording accuracy and resolution
  - Station history inspections and surveys of reference gauges
  - Equipment calibration history and any other information affecting data quality

4. DISCHARGE MEASUREMENT

The previous section described how to install and maintain pressure transducers to measure continuous stage (water level). Taken alone, stage measurements provide information about streamflow patterns, including the timing, frequency, and duration of high flows (Roy et al. 2005). Stage data alone do not provide quantitative information about the magnitude of streamflows or flow volume, which can complicate comparisons between streams (U.S. Environmental Protection Agency (USEPA) 2014). In addition, channel shape can change over time, such that a similar stage measured at a given location during two separate years could represent different flows (USEPA 2014). This change would not be captured with stage data alone. Therefore, to assess patterns and changes in stream hydrology, it is valuable to convert stage measurements into streamflow (USEPA 2014). The most common approach for converting stage to discharge is to develop a stage-discharge rating curve. To develop a rating curve, a series of discharge (streamflow) measurements is made at a variety of stages, over as wide a range of flows as possible.

This section describes how to take discharge measurements in wadeable streams. This involves measuring the depth and velocity of water passing through several segments along a given cross-section of stream. The water depth and the measurement intervals across the stream are obtained using a rod for depth and a survey tape for distance. A current meter is used to measure the stream velocity at each interval. Each measured velocity is multiplied by its contributing flow area and the resulting flows are summed across the cross-section to provide a total flow.

4.1 Measurement Frequency

To establish a rating curve at a new site, five to ten discharge measurements should be taken over as wide a range of flows as possible (USEPA 2014). Once a rating curve is established, it is recommended to measure discharge at least once annually, and if possible, also after large storms (or any other potentially channel-disturbing activities) to verify or update the curve (USEPA 2014). If new measurements are more than 15% off the rating curve, follow-up measurements should be taken to determine whether a shift has occurred and (if necessary), to establish a new rating curve (USEPA 2014).

4.2 Equipment

The basic equipment required to perform discharge measurements in wadeable streams includes:

- **Current meters** measure point velocity. There are several types of current meters available on the market. They are grouped into three main categories: mechanical current meters, electromagnetic current meters, and acoustic doppler velocity (ADV) meters. These are described in Appendix D.
- **Top-setting wading rods** are used to measure water depth at verticals and to set the current meter at the appropriate depth. The design incorporates a graduated stainless steel hexagonal rod and a parallel round aluminum rod, connected by an aluminum handle. The current meter is attached to a sliding support on the base of the aluminum rod, which is allowed to slide vertically for rapid positioning of the meter. A vernier on the handle permits automatic setting of the meter to any desired depth (i.e., 0.6, 0.2, or 0.8 depth) (Figure 3).
- **A measuring tape and stakes** are used to define the specific location of the cross-section at which depth and velocity are measured.

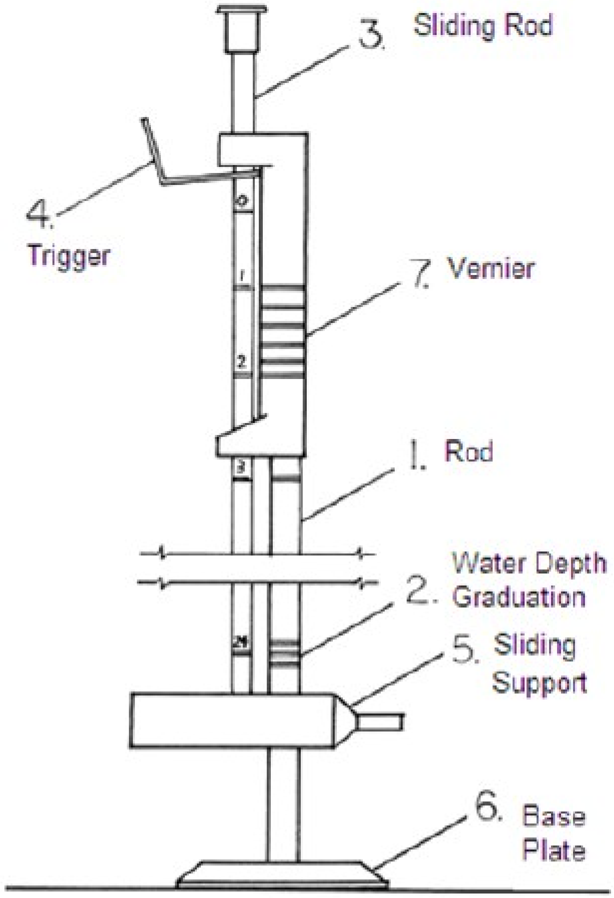

Figure 3: Components of a top setting wading rod (Source: RISC 2018)

4.3 Site Selection

The site (metering cross-section) should be selected based on the following characteristics:

- The metering section should be confined to a single channel for all stages (i.e., no flow bypassing the metering section).
- The metering section should be perpendicular to the general direction of flow. A procedure for determining angle of flow corrections is described in Section 4.6.
- The metering section should be located where the bed and banks of the watercourse are straight and uniform:
  - Upstream - for a distance of approximately five times the width of the metering section.
  - Downstream - for a distance of approximately twice the width of the metering section.
- The channel bed at the metering section should be as uniform as possible. It should be free from vegetation, large boulders and obstructions such as bridge piers.
- One or more locations at which the full range of flows can be measured with available equipment. For flows that cannot be waded, it is more economical to use an existing bridge or culvert for high flow measurements.
- The metering section normally should accommodate 20 verticals/subsections of uniform discharge, and flow in any one subsection or panel should not be more than 10% of the total flow. However, measurement points should not be closer than 0.15 m when a Price Type AA meter is used. This factor should be considered during site selection.
- The spacing of verticals along the metering section is not usually uniform. Where the water is shallow and/or slow moving, the spacing will be greater than where the water is deep and swift.
- If possible, a manual gauge or reference gauge should be located in the same metred section.
- The metering section should be within reasonable proximity to the gauge.
- No inflows or outflows between the gauging point and the measurement section. If this is unavoidable, an auxiliary station will be required to gauge these flows.
- Both the low flow and high flow measurement sections should be clear of aquatic vegetation.

When selecting a metering cross-section, consider potential changes to the stream following restoration actions. For example, if monitoring the impacts of a floodplain reconnection project on streamflow conditions, the metering section should be within a single channel located downstream of all reasonably anticipated geomorphic changes. Attempting to measure discharge in an anastomosing or braided channel presents additional challenges for developing a stage-discharge rating curve (Bullard et al. 2007). However it is still valid to obtain supplementary stage, discharge, and/or velocity information from within a multi-channel reach, and this may be done with stage loggers, current meters, or remote sensing techniques such as Large-Scale Particle Image Velocimetry (LSPIV; Saletti 2024).

4.4 Measuring by Wading

Wading measurements are preferred to those obtained by other means as they are relatively easy to carry out and computing the discharge can be more straightforward than other techniques. However, in very small watercourses (e.g., ditches), the presence of the technician in the water may significantly affect flow. In these instances, the technician should stand on a plank or log placed across the watercourse. The step-by-step protocol for conducting discharge measurements by wading is outlined below.

Step 1 – Locate the metering section

- During the initial site reconnaissance and selection, the technician would have assessed the channel to select locations to carry out discharge measurements (Sections 2.3 and 4.3). The location of the metering section may vary with changes in stage or channel conditions, but ideally, once the metering sections for low, medium, and high flow measurements have been selected, they should not be changed.
- The location of the metering section should be well-defined (e.g., iron pin or rebar driven in to the ground above the high water mark, normally on the right bank). This point is used as the initial point and should be referenced to another permanent feature near the metering section or gauging station.
  - Note, these data are often required for detailed studies long after the gauging station has been established or discontinued, and is particularly important where it is desirable to define channel erosion or deposition.

Step 2 – Set up the tagline and establish verticals

- Wade across the metering section to gather an overall impression of the channel depths and velocities. Look for rocks and debris that might be removed from the channel bed to improve the metering section. Be certain (particularly for very small watercourses) that removing rocks will not affect or alter the control.
- Anchor the tagline (i.e., measuring tape) with the zero referenced to the initial point (described in Step 1).
- Wade across the watercourse and position the tagline at a right angle to the direction of the current and secure it to the shore. Note the overall width of the metering section.
- Determine the approximate spacing of the verticals according to the flow pattern, considering the following:
  - The metering section should accommodate at least 20 verticals/subsections (not including the water's edge), and flow in each subsection or panel should not be more than 10% of the total discharge. Note, measurement points should not be closer than 0.15 m when a Price Type AA meter is used. This factor should be considered during site selection.
  - The spacing of verticals along the metering section is not usually uniform. Measurement points should be closer together in portions of the cross section where flow is more concentrated and depths are greatest, in order to ensure that no more than 10% of the total discharge passes through any single segment. Measurements may be farther apart where the flow is lowest and depths are shallow.

Step 3 – Measure Discharge

- Record the starting time and gauge height reading on the 'Discharge Measurement Field Form' (Appendix E).
  - Note, an accurate determination of the mean gauge height is essential for plotting the results of the discharge measurement. If the stage appears to change while the measurement is in progress, it is necessary to obtain additional readings during the progress of the measurement, if safe to continue.
- Record the tagline distance for the edge of water. If there is a steep drop at the edge of the stream, the first 'vertical' depth and velocity observation should be taken close to the edge.
- Move to the next vertical (based on the spacing established in Step 2). Record the distance indicated by the numbered marker on the tagline. Observe and record the water depth.

- Before starting the velocity measurement, the averaging velocity period in the instrument should be set to at least 40 seconds and preferably to a range between 50 to 70 seconds
- Set the current meter to the correct depth using the vernier (based on the criteria below)
  - Where water depth in the vertical is < 0.75 m, measure and record the velocity at 60% of the total depth (i.e., 0.6 x depth, as measured from the water surface; one-point method).
  - Where water depth in the vertical is > 0.75 m, measure and record the velocities at 20% and 80% of the total depth (two-point method).
  - When streambeds are very rough, irregular or covered with aquatic growth, the 0.2 and 0.8 method is not entirely satisfactory. In such conditions the average of velocities measured at 0.2 and 0.8 depths is averaged with the velocity observed at the 0.6 depth (three-point method). This can be expressed with the following equation: $Vmean = (V_{0.2} + V_{0.8})/4 + (V_{0.6})/2$
- With the current meter set to the correct depth, measure the mean water column velocity.
  - Hold the wading rod so that the sensor is pointing upstream and perpendicular to the tag line. If the direction of flow is unclear, rotate the rod slightly to find the angle at which the maximum velocity is observed.
  - Start the velocity measurement and hold the current meter steady for the duration of the measurement until the velocity value appears on the display. Record the displayed value on the data sheet.
  - Note, the technician should stand to the side and downstream from the meter so as not to influence the velocity. Other field personnel should avoid in-water activities on the upstream side of the metering section during the discharge survey, as this can affect the flow readings.
- Repeat this procedure at each vertical until the watercourse is traversed and the measurement is completed.
- After completing the measurement, note and record the time and gauge height reading.

Precautions and tips to obtain accurate measurements by wading

To obtain accurate measurements by wading, the technician must pay attention to detail and technique. The following steps should be implemented where appropriate:

- **Correct positioning of the tagline** – the tagline should be placed across the channel, perpendicular to the direction of the current. In instances where angular flow occurs, the cosine of the horizontal angle should be recorded.
- **Improving the metering section** – Where necessary, take the time to improve the metering section by removing boulders and debris from both the metering section and the area immediately above it. Remove weeds for a distance of about three times the depth from the area upstream and downstream from the section. After the modifications are made, allow for sufficient time for conditions to stabilize before proceeding with the measurement. Note that all improvements to the metering section should be completed before starting the measurement (i.e,. **do not** make changes to the metering section (such as moving rocks) during the course of the discharge measurement).
- **Spacing of the verticals** – Obtain 20-25 observations of both depth and velocity for one complete measurement. For very narrow cross-sections this will require the verticals to be spaced more closely.
- **Position of the current meter** – The wading rod should be held in a vertical position, with the current meter oriented perpendicular to the tag line while making the velocity observation.

- **Velocity observations** – If depths are sufficient (>0.75 m), the 0.2 and 0.8 method should be used for observing velocities. To set the current meter on 0.2 depth position (or 0.8 up from streambed) on the wading rod, simply double the value of the observed depth, and for 0.8 depth position (or 0.2 up from streambed) one-half of the observed depth on the vernier setting of the top setting wading rod. For example, if the observed depth is 1.0 m then to get 0.2 depth, set 2.0 on the rod, and for the 0.8 depth, set 0.50 on the rod.
- **Uneven channel bed** – If the channel bed is rough, care should be taken to adjust the observed depths so that they reflect both the tops of the boulders and the depths between them. Measuring verticals should be equidistant around the vertical line that defines the break point on the edge of the submerged obstruction. Depth and velocity observations are made just before and just after the point where the change takes place.

4.5 Measuring from a Bridge

If a watercourse cannot be waded, discharge measurements can be made from either the upstream or downstream side of a bridge. A bridge rod can be used to position the current meter in the watercourse. Factors to consider when selecting where to take discharge measurements include:

- Physical conditions at the bridge, such as location of the walkway, traffic hazards, accumulation of debris on pilings or piers.
- Advantages of measuring from the upstream side of the bridge:
  - Hydraulic characteristics on the upstream side of bridges are usually more favourable
  - Drift material can be seen coming downstream and therefore avoided more easily
  - The channel bed at the upstream side of the bridge is not likely to be scoured as badly as the downstream side
- Advantages of using the downstream side of the bridge:
  - Bridge rods are less likely to become damaged from bending over the edge of the bridge if caught by the current and/or debris
  - The vertical angle of a cable-suspended current meter is more easily observed.
  - Bridge abutments and piers can straighten flow lines in some cases
  - If the bridge is angled across the channel, a single horizontal correction for angular flow can be applied to the measured discharge

The step-by-step protocol for conducting discharge measurements from a bridge is outlined below.

Step 1 - Assemble the bridge rod and current meter

- Assemble the bridge rod, using enough sections to reach the deepest point of the channel bed. Probe the channel bed with the rod to determine the range of depths along the metering section. Determine the number of outer rod sections required, based on the following criteria:
  - To determine the depth, raise the meter to the surface, and read the graduated inner rod where it emerges at the top of the location device. From this value, subtract the number of outer sections (1 m per section) to obtain the depth.
  - Multiply the depth by 0.2 and lower the meter to the obtained value. This is done by subtracting the number of metric sections of outer rod.
  - The two values above represent the maximum range of readings to be made from the bridge deck. They also determine the number of inner and outer rod sections required.
- Assemble the current meter

Step 2 – Set up the tagline and establish verticals

- Extend the tagline along the top of the bridge rail.
- Locate the initial point. It will usually be the bridge abutment and should be on the right bank.
- Prepare the discharge measurement field notes. Based on the initial exploration of depth and velocity distribution in Step 1, decide on the spacing of the verticals and the mode of measurement required (e.g., a 0.6 depth measurement if depths in the panel < 0.75 m).

Step 3 – Measure discharge: Mid-section method

- Record the starting time and gauge height reading on the 'Discharge Measurement Field Form' (Appendix E).
  - Note, an accurate determination of the mean gauge height is essential for plotting the results of the discharge measurement. If the stage appears to change while the measurement is in progress, it is necessary to obtain additional readings during the progress of the measurement.
- Record the tagline distance from the initial point to the edge of the water. Record the depth at water's edge.
- Position the rod at the first vertical. Position the meter at the water surface with the propeller or cups half submerged. Lock the locating device and read the graduated rod at the top of the outer rod. Subtract the number of 1-m sections of outer rod and record distance from the initial point (station) and depth.
- Calculate the meter depth setting(s).
- Set the current meter to the required depth and allow it to stabilize to obtain the velocity.
- Obtain the velocity by recording the observed velocity averaged over 40-70 seconds.
- Continue measuring distance, depth, and velocity along the cross-section.
- End the measurement at the water's edge on the left bank. Record the distance from initial point (i.e., tagline distance), depth, time, and water level.

4.6 Errors Affecting Measurement Accuracy

The section below outlines factors that can contribute to inaccuracies when observing widths, depths, and velocities.

- Width Measurement – Wind can oscillate the tape and lead to movement of the tape anchorage points and/or mistakes in observing values.
- Depth Measurement – Depth observations made by rod are subject to random errors such as the sinking of the rod into a soft streambed, failure to identify obstructions in the cross-section between soundings, and incorrect reading of the graduated rod.
- Measurement of Velocity – potential errors related to the measurement of velocity includes calibration of current meters, the direction of flow, the duration of the observation time, and the number of observation verticals as well as the number of observation points in each vertical.
  - Calibration – defined as the process of comparing the response of a measuring device with a calibrator or a measuring standard over a range. All individual meters must be calibrated as per manufacturer's specifications.
  - Field Verification - defined as the act of confirming the equipment accuracy and specifications using a traceable standard or master. Every meter must be verified at least once in a year. Verification of the meter can be done by comparison to the readings of a

recently calibrated industry standard current meter such as Price Type AA, ADV meter, etc. or by following manufacturer's instructions.

- Direction of flow - Discharge measurement cross-sections are usually chosen so that the flow is perpendicular to the cross-section. Even though they are carefully selected, it is not always possible to avoid oblique flows at some of the verticals. At these verticals the velocities must be corrected by applying an appropriate cosine coefficient (note: FlowTracker does this correction automatically). Random errors may be introduced when observing the angle of flow if it is assumed that the angle observed at or near the surface remains the same throughout the entire depth. Other sources of error can be introduced when using rod suspensions and in particular when used in deep fast-flowing water. Although the current meter can be misaligned both vertically and horizontally with the direction of flows, the most significant error will result from vertical misalignment. The current meter will under-register if tilted above or below the horizontal, and the magnitude of the error will depend upon both the velocity of the water and the angle of departure.
- Duration of observation time - Pulsations in velocity are evident in all streams. Because pulsations are random in nature, the effects of pulsation will be eliminated when velocities are observed for a sufficient length of time. It is recommended to observe the velocities for 40 – 70 seconds to eliminate the pulsation effect.
- Number of observation verticals – errors will be introduced if the numbers of observations made to define the cross-section are not sufficient. This can be overcome by obtaining additional depth observations.
- Distribution of observation verticals – The verticals should be spaced so that the velocities observed are more representative of those in the preceding half panel and the following half panel. The use of 20-25 verticals is considered optimum. Each subsection should not be more than 10% of the total flow and when setting up the vertical interval the operator should target 5%.
- Number of observation points in a vertical – The mean velocity in a vertical is normally obtained by measuring at one or two points in that vertical. When depths are less than 0.75 m, the one-point method (0.6 depths) should be used.

Overall, errors in the measurement of width, depth, and velocity as well as the lack of care in choosing the number of verticals and observations in a vertical, all combine to reduce the overall accuracy of a discharge measurement. These errors can be reduced if the technician obtaining the measurement can recognize the potential problem areas and take the appropriate precautionary measures to avoid or minimize them.

5. DEVELOPMENT OF A STAGE-DISCHARGE RATING CURVE

A stage-discharge rating curve is produced by plotting instantaneous flow measurements and stage heights. Refer to ECCC (2016) for detailed guidance on the development and maintenance of stage-discharge models.

Simple rating curves can be developed using standard software packages such as Microsoft Excel. However, it is becoming increasingly common to use specialized software designed specifically for hydrometric data analysis (e.g., Aquarius).

Alternatively, the Hydrometric Rating Application (HydRA) is a platform (available at: https://hydra.nrs.gov.bc.ca/rctool/) that allows users to upload stage and discharge datasets, develop and optimize rating models as well as save and compare results from previous sessions. Refer to the webpage directly (https://hydra.nrs.gov.bc.ca/rctool/) for detailed guidance on how to use HydRA features for rating curve development.

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APPENDIX A: HYDROMETRIC SITE VISIT

This section describes the activities that field technicians should complete prior to and during a typical site visit when first establishing hydrometric monitoring stations and when completing repeat visits to the station. It is intended as a stand-alone, step-by-step guide to support a systematic and repeatable workflow and presents information specific to field procedures.

A.1 Station Visit Preparation

Before departing for the site, technicians should complete a thorough review of the following:

- Review and print relevant documents detailing the latest station information including:
  - benchmark names, locations, and elevations
  - Reference gauge locations and elevations
  - Notes from previous field visit. Ensure that any issues encountered on previous visits have been addressed, and the field team is equipped to handle any further issues that may arise.
- Review the historical time series or a nearby and relevant real-time Water Survey of Canada monitoring station to estimate expected discharge at the site (i.e., increasing, stable, or decreasing). Consider the equipment necessary to perform measurements at the site based on the anticipated stream conditions (high, medium or low flows, ice etc.).
- Review weather forecasts to determine whether discharge may be likely to change during the site visit and prepare accordingly.
- Ensure all field equipment is fully charged and functioning optimally, including:
  - Current meters (check condition and calibration)
  - Communications equipment (e.g., radio, satellite phone, inReach etc.)
  - Software and cabling
  - Laptops
  - Backup supply of batteries, data loggers, pressure sensors etc.
  - Spare instrumentation
- Obtain all relevant permits and/or permissions (e.g., private land owners)

Table A-1: Equipment Checklist – General Items

Waterproof field notebook and pencils	☐
Personal floatation device (PFD)	☐
Throw bag	☐
Field forms printed on waterproof paper and/or field tablets	☐
Charging cables for electronics	☐
External battery packs	☐
GPS	☐
Chest waders	☐

Communication devices (radio, satellite phone, inReach etc.)	☐
Digital Camera	☐
Calculator (optional)	☐
Wildlife deterrents (bear spray, air horns etc.)	☐
Safety glasses (to use while operating power tools etc.)	☐
Duct tape	☐
Tuck tape	☐
Electrical tape	☐

A.1.1 Site Safety Assessment

Upon arrival to the site, visually scan the area and note/discuss safety hazards with all members of the field team. Hazards to consider while working around water include, but are not limited to:

- Surprise wildlife encounter due to reduced visibility and stream noise
- Debris floating downstream (e.g., ice, woody debris, etc.)
- Rocks or mud that could cause foot entrapment
- Unstable banks and uneven footing
- Deep water that could swamp chest waders
- Swift moving water that can cause a fall
- Hypothermia caused by falling in cold water

Field teams should discuss and implement measures to reduce the risks associated with working in and around water:

- Check the upstream and downstream work areas for hazards
- Perform regular wildlife scans and have deterrents on-hand
- Wear PFDs when conducting work in or around water
- While wading in water, consider using a staff or top-setting rod to help balance
- Have a throw bag ready for use
- Bring spare clothing as necessary
- Crew members not working in the water should watch upstream for incoming debris and be prepared to assist the team working in the water as necessary.

A.2 Station Installation

A.2.1 Sensor Configuration and Launch

Start the pressure transducer before deployment to ensure it is functioning properly and to configure the logging session. To do this, connect the pressure transducer to a laptop and use the appropriate software (specific to the type of sensor used) to communicate with it. Refer to the manufacturer's instructions for software-specific information. Note, this should be done in the office **before** leaving for site. <u>If using a non-vented pressure transducer (i.e., requires external compensation for changes in barometric pressure via a barometric sensor deployed nearby on land) configure both sensors so that recordings are taken at the same time.</u>

In general, the steps to configure and start the logging session are as follows:

1. Connect the USB connector to the pressure transducer and plug it in to a laptop (with the appropriate software/drivers installed).
2. Open the software and select the appropriate COM port (it will likely be USB)
3. Make sure your computer clock is correct, then synchronize the logger clock with your computer clock.
4. Name the session ID (e.g., SiteName_Year_LoggerType)
5. Set the logging interval to 15 minutes, starting at the top of the hour. Note, site specific flow regimes may necessitate a shorter interval.
6. Configure the memory mode based on the project requirements. This may involve setting the number of records recorded (e.g., 100,000), or selecting between different memory modes (continuous versus slate mode). Read the manufacturer's instructions carefully to determine how to configure the memory settings for the type of sensor used.
7. Check that the session is configured correctly, then select Start.

A.2.2 Benchmark Installation

A list of equipment needed for benchmark installation is provided in Table A-2.

Table A-2: Equipment Checklist – Benchmark Installation

Safety glasses (to use while operating power tools etc.)	☐
Lag bolts (for benchmarks in trees)	☐
4 to 6 expanding anchor bolts with nuts and washers (4" – 3/8" rock bolts for benchmarks)	☐
Cordless electric drill with assorted bits	☐
Hammer or rotary drill (e.g., Hilti rock drill) for mounting to rock)	☐
Multiple 3/8", 7" long drill bit; for use with Hilti rock drill (or other hammer/rotary drill)	☐
Hammer	☐

Aluminum tags (numbered; for marking benchmarks)	☐
Flagging tape	☐
Spray paint (bright colour for marking benchmarks)	☐
Flexible, galvanized metal wire (useful for securing aluminium tags to benchmarks)	☐

1. Identify suitable sites for the installation of three independent, readily identifiable benchmarks. Site selection considerations include:

   a. Sites should provide a high degree of stability (e.g., bedrock, large immovable boulder, fixed structure etc.)

   b. Benchmarks should be spread out, away from the river bank and above the maximum expected water level.

2. Use a hammer drill or rotary hammer with a concrete drill bit to drill holes in the marked locations.
3. Pound concrete wedge anchors (bolts) into the holes using a hammer until secure.
4. Affix the unique numbered aluminum tag to the bolt using metal wire and secure in place with the washer and nut.
5. In addition to the unique numbered aluminum tag, use flagging tape and spray paint to clearly mark the location of the benchmark.
6. Once benchmarks are established, sketch their location on the data sheet (Appendix E), clearly documenting both benchmarks and relevant site features that can be used to locate the benchmarks in subsequent surveys.
7. Take pictures of the benchmarks with clear identifying features.
8. Complete a benchmark levelling survey to establish gauge datum (Refer to Section A.2.6 below)

A.2.3 Instream Pressure Transducer Deployment

Instream pressure transducers should be housed within a deployment structure to protect the sensor from moving debris and sediment and allow free flow of steam water across the tube and sensor. Inexpensive housings can be constructed from 1.5 inch diameter Schedule 40 (or stronger) PVC pipe (Figure A-1, USEPA 2014). Multiple holes (½ inch or larger) should be drilled into the PVC to allow water to fluctuate at the same rate as in the stream. The housings can be painted black to make them less visible if vandalism is a concern.

Figure A-1: Example of a protective housing for a vented pressure transducer made of PVC pipe (Source: USEPA 2014)

Once housed in a protective casing (e.g., PVC pipe with drilled holes), pressure transducers should be installed as deep in the channel as possible to allow for continuous monitoring of water levels at all range of flows. An example of an installation technique is provided below (adapted from USEPA 2014), though site-specific conditions or restoration monitoring objectives may necessitate alternative approaches. Typically, non-vented pressure transducers will need to be removed from the stream to download data. The method presented below facilitates easy removal and re-deployment of the pressure transducer without disturbing the deployment structure. A list of equipment needed for pressure transducer installation is provided in Table A-3.

Table A-3: Equipment Checklist – Pressure Transducer Installation

All Sites	
Safety glasses (to use while operating power tools etc.)	☐
PVC pipe for transducer (drilled with 1/2-in or larger holes)	☐
PVC pipe for data logger or barometric pressure logger	☐
Galvanized or stainless steel conduit straps, hangars, and hose clamps	☐
Heavy duty zip ties/cable ties	☐
Screws (stainless steel or brass)	☐
Screwdriver	☐
Drill and assorted drill bits for wood and PVC	☐
Level	☐
Hammer or rotary drill	☐
Concrete drill bit	☐

Concrete wedge anchors with nuts, and washers (stainless steel or galvanized)	☐
Hammer	☐
Hack saw (to cut PVC pipe/conduit if needed)	☐
Pressure transducer (and associated instrumentation)	☐
Barometric pressure sensor (if using non-vented pressure transducer)	☐
Field laptop with USB connector cable and appropriate software installed	☐
If using a non-vented transducer	
Non-stretch cable or wire	☐
Wire rope clamps	☐
Long bolt and wing nut	☐
Extra-long PVC (should extend out of water in high flows)	☐
PVC with caps for barometric transducer	☐
If using a vented transducer	
Garden staples	☐
PVC pipe for data logger	☐
PVC cap or locking well cap	☐
Stainless steel conduit straps	☐
Long lag screws	☐
Wire cable	☐
Replacement desiccant (if necessary)	☐

1. Use a PVC pipe that is approximately the same height as the gauge board.
2. Attach a non-stretch cable or rope (e.g., coated stainless steel cable) to the transducer and make a loop at the non-transducer end, secured with wire rope clamps (Figure A-2). The cable should be long enough that the transducer will always be under water but not so long that it will come in contact with the streambed.
3. Place the looped cable over a long bolt with a wing nut that runs through the top of the PVC (Figure A-2). Clearly mark the holes for the bolt so that the transducer is placed back at the same elevation every time it is removed.
4. Install the transducer/housing on a fixed object (e.g., bridge abutment, boulder, anchored staff gauge board):
    i. Install the transducer close to the staff gauge so that stage changes are comparable.

ii. Use concrete anchors and conduit straps, hangars, or hose clamps to attach the PVC vertically to a bridge or boulder.

iii. Ensure that the PVC pipe and transducer are installed slightly above the streambed to reduce chances of fouling by fine sediments.

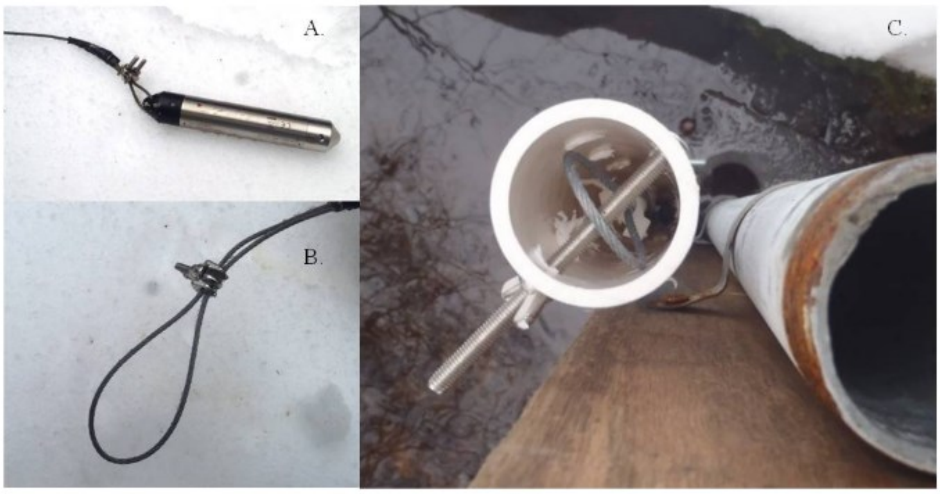


Figure A-2: Non-vented pressure transducer installation on a staff gauge board. (A) non-stretch cable is attached to the transducer, (B) a loop is made at the non-transducer end of the cable, (C) a cable with transducer is suspended over the bolt from the top of the PVC pipe (Source: USEPA 2014)

A.2.4 Deployment of On-land Components

Both vented and non-vented transducers have components that must be installed on land. If the transducer is vented, it has a vented cable that connects to a data logger on land. While, if the transducer is non-vented, barometric pressure readings must be obtained from a separate device that is located nearby on land.

The general installation approach is as follows (Adapted from USEPA 2014):

1. Identify a suitable location on land above where high flows or the snow pack will reach. Trees are generally the most suitable attachment points for these devices.
2. Attach a PVC pipe to a fixed object on the bank. For example, the pipe can be attached to a tree with U-shaped conduit straps and long screws or zip ties (Figure A-3). Adjust the length of the pipe to adequately cover the device with some extra room.
3. Place the device within the PVC pipe.
4. Place a PVC cap or locking well cap on top of the PVC pipe to protect the equipment and discourage vandalism.

Specific considerations for the data logger include:

- For vented transducers with excess cable, coil the cable and zip tie it to the tree or PVC pipe.
- Use garden staples (4 inch U-shaped steel staples), rocks, and leaf litter to hide the data logger cable. For added protection, the cable can be encased in a PVC pipe (or flexible aluminium conduit).

Specific considerations for the barometric pressure sensor include:

- Place the device on land as close as possible to the instream pressure transducer. In instances where multiple non-vented pressure transducers are deployed close together (i.e., less than 30 km distance and within 300 m elevation, Solinst Canada Ltd. n.d.), one barometric pressure sensor can be used to compensate water level data collected at stations within close range.

- The PVC housing should be drilled with holes to facilitate air flow (Figure A-4).
- Holes should also be drilled in the bottom of the PVC housing to prevent condensation and laterally blown rain and snow from filling the cup.

Figure A-3: Vented transducer data logger installation. The transducer is attached to the tree with stainless steel conduit straps (Source: USEPA 2014).

Figure A-4: Barometric pressure sensor installation for a non-vented transducer (Source: USEPA 2014)

A.2.5 Staff Gauge Installation

If a staff gauge is to be installed, it can be attached to a fixed object in the stream (e.g., boulder, bridge, weir; with appropriate permissions) or installed in the streambed (Figure A-5). In general, attaching gauges to a fixed structure is preferable, as this minimizes the chance that the gauge will shift in high flows (USEPA 2014). Staff gauges are typically available in 1 m sections, though multiple sections can be combined (stacked vertically) to encompass the full range of stages expected to occur within the stream. The staff gauge plate(s) should be attached to a wood backing board that is at least as high and wide as the staff gauge plate(s) and able to withstand being submerged in water. A list of equipment needed for staff gauge installation is provided in Table A-4.

Table A-4: Equipment Checklist – Staff Gauge Installation

All Sites	
Safety glasses (to use while operating power tools etc.)	☐
Staff gauge plate(s) and wood backing board	☐
Screws (stainless steel or brass)	☐
Screwdriver	☐
Assorted drill bits for wood	☐
Bubble Level	☐
Step ladder (if installing more than one section of a staff gauge)	☐
Wildlife cameras (optional)	☐
For installation to a fixed structure	
Concrete wedge anchors with washers and nuts (stainless steel or galvanized)	☐
Hammer or rotary drill	☐
Concrete drill bits (multiple)	☐
Hammer	
For installation into the streambed	
Post pounder and/or sledge hammer	☐
Galvanized or stainless steel pole with cap, or angle iron	☐
Galvanized or stainless steel conduit straps, hangars, and hose clamps	☐
Nut driver set (if using hose clamps)	☐
Pry bar	☐

The general procedure for installing a staff gauge and backing board to a **fixed structure** is as follows (adapted from USEPA 2014):

1. Screw staff gauge plate to wood backing board (this can be done prior to going in to the field).
   a. A staff gauge should be installed such that it can be read at all water levels. If a site at high water is very deep, two staff gauges may be stacked to ensure reading is possible at all flows.
2. Identify location for staff gauge installation close to the pressure transducer installation.

3. Level the gauge board on the fixed object and mark locations for concrete wedge anchors. Use concrete anchors on both sides of the board and at varying elevations to ensure the board's stability. Two anchors are typically sufficient for a 1 m gauge and three for a 2 m gauge.
4. Use a hammer drill or rotary hammer with a concrete drill bit to drill holes in the marked locations.
5. Pound concrete wedge anchors (bolts) into the holes using a hammer until they are secure.
6. Drill holes in the gauge board for concrete wedge anchors. Place the board over the bolts and screw it into place using the nut
7. Once installed, read water level to the nearest mm by reading the bottom of the meniscus. Record the reading and the time. Take photos and/or videos of the staff gauge.
8. Optionally, install a trail camera to face the staff gauge plate. Timelapse images of water level can be used to validate pressure transducer readings and provide additional context. Trail camera imagery may also be used to provide qualitative (or quasi-quantitative, see Bellucci *et al.* 2020, Zargarpour and Franklin 2025) information on flow and fish passage when water levels are too low to be detected by a pressure transducer.

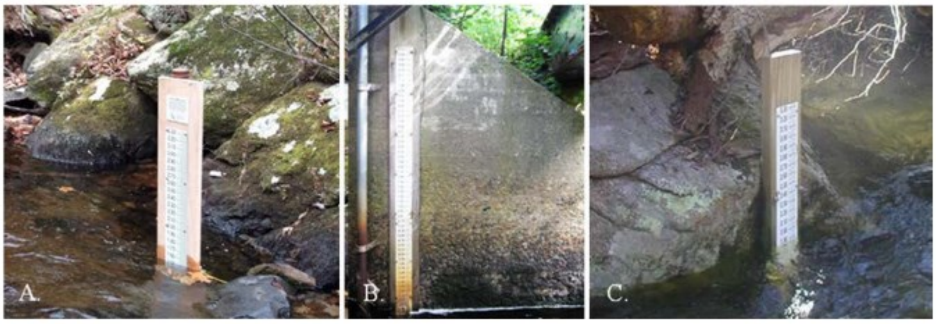


Figure A-5: Examples of staff gauge installation techniques (A) streambed (B) fixed object - bridge wing wall, (C) fixed object – boulder (Source USEPA 2014)

The general procedure for installing a staff gauge and backing board into the **streambed** is as follows (adapted from USEPA 2014):

1. Screw staff gauge plate to wood backing board (this can be done prior to going in to the field). A staff gauge should be installed such that it can be read at all water levels. If a site at high water is very deep, two staff gauges may be stacked to ensure reading is possible at all flows.
2. Identify location for staff gauge installation close to the pressure transducer installation.
3. Use a pry bar to test the streambed in the pool for locations where the pole can be driven into the streambed.
4. Drive a galvanized steel pipe with cap (or angle iron) into the streambed using a post driver or sledgehammer. Poles of 6 feet and 9 feet are generally sufficient for 1 m and 2 m gauges, respectively. The pole should be driven into the streambed at least 1 m or until it is very stable and unlikely to be knocked over in high flows. The pole should be as straight as possible so that when the board and gauge are attached, they can be levelled. A step ladder can be helpful for driving the pole.
5. Attach the combined board/gauge to the pole using galvanized or stainless steel conduit straps (use at least two for a 1 m gauge). The board should be parallel to the flow and, if possible, positioned so that it can be read from the stream banks.
6. If a tree is nearby on the stream bank, a metal bracket or brace can be attached from the gauge to the tree to increase stability.

7. Once installed, read water level to the nearest mm by reading the bottom of the meniscus. Record the reading and the time. Take photos and/or videos of the staff gauge.
8. Optionally, install a trail camera to face the staff gauge plate. Timelapse images of water level can be used to verify pressure transducer readings and provide additional context for stage measurements.

A.2.6 Levelling Survey

A list of equipment needed for levelling surveys is provided in Table A-5.

Table A-5: Equipment Checklist – Levelling Survey

Engineer's level	☐
Stadia rod and rod level	☐
Tripod	☐

1. Set up the level on stable ground and ensure it is properly levelled (the levelling bubble must be completely within the setting circle).
2. Identify one benchmark as the primary reference point (i.e., primary benchmark). This benchmark will be assigned an arbitrary local elevation of 100.000 m above the datum.
3. Survey the benchmarks, top of the staff gauge plate (if installed), and water level in sequence, beginning with the primary benchmark and ending on a Turning Point. Note, the water level should be surveyed at a point representative of the water level the pressure transducer is reading.
4. While keeping the stadia rod fixed to the turning point, reposition the level and re-survey the turning point, water level, top of staff gauge plate (if installed), and benchmarks, beginning at the turning point and ending at the primary benchmark.
5. Check addition and subtraction to ensure the elevations measured in the first survey transect are the same as the second survey transect.

A working example is provided below:

The bench mark identification number is entered in the column headed Station. On the top line, in the column headed BS (backsight), the reading obtained with the levelling rod held on the bench mark or point of known elevation is entered as the backsight. The value for the column HI (height of instrument) is computed by adding the backsight value to the known bench mark elevation (i.e., 0.046 + 100.000 = 100.046).

One line down, in the next column headed foresight (FS), the foresight reading for the point for which an elevation is to be determined is observed and recorded. This value is then subtracted from the HI and the result is the elevation of the foresight point. This is entered in the elevation column and on the same line as the foresight just observed. (i.e., 100.046 - 0.282 = 99.764).

When the instrument is moved, the new height of instrument is determined by a backsight on the Turning Point (TP; i.e., 98.688 + 1.401 = 100.089). The observation and notes are continued in this manner until the circuit is closed by levelling back to the original starting Station.

Station	BS (+)	HI	FS(-)	Elevation	Notes
BM 001	0.046	100.046		**100.000**	Primary BM
BM 002			0.282	99.764	
BM 003			0.200	99.846	
Water Level 1			1.524	98.522	
Water Level 2			1.525	98.521	
Turning Point			1.358	98.688	Instrument is moved after foresight is taken on the TP
Turning Point 2	1.401	100.089			
Water Level 2			1.568	98.521	
Water Level 1			1.566	98.523	
BM 003			0.242	99.847	
BM 002			0.325	99.764	
BM 001			0.088	100.001	Primary BM

A.2.7 Discharge Measurement

A list of equipment needed for discharge measurements is provided in Table A-6.

Table A-6: Equipment Checklist – Discharge Measurements

Current Meter	☐
Top-setting wading rod	☐
2 pieces of rebar for securing measuring tape (optional)	☐
Measuring tape	☐

Select Location

Consider the following criteria when deciding where to conduct manual stream discharge measurements. The location may change between visits depending on the channel conditions at different water levels. Ideally, once the metering sections for low, medium, and high flow measurements have been selected, they should not be changed.

- The transect should be safe to cross and near the hydrometric monitoring station
- The streambed should be as flat as possible and free of nearby upstream obstructions (e.g., large boulders, thick instream vegetation etc.)

- The transect should be relatively straight with even, non-turbulent flow where possible.

Measuring Tape Setup

- Run the measuring tape across the width of the stream from one stream bank to the other, perpendicular to the stream flow. For channels less than 3 m in width, the stadia rod can be used as a tape measure.
- Secure the measuring tape with rebar or tie off to a tree to keep the tape taut and prevent it from dipping into the water or coming loose. The transect must cover all channels of a braided stream.

Determine Measurement Interval

- Measure the wetted width of the stream (using the measuring tape that has been secured across the width of the channel) and determine an appropriate measurement interval. The goal is to obtain a **minimum** of 20 measurements across the stream channel.
- The spacing of verticals along the metering section is not usually uniform. Measurement points should be closer together in portions of the cross section where flow is more concentrated and depths are greatest, in order to ensure that no more than 10% of the total discharge passes through any single segment. Measurements may be farther apart where the flow is lowest and depths are shallow.
- Flow in each subsection or panel should not be more than 10% of the total flow for the stream channel. If any measurements exceed 10% of the flow (current meters will often display this information), take an additional measurement next to the measurement exceeding 10% and between existing measurement locations.

Measure Velocity and Depth

- Beginning at either wetted stream bank, record the distance measurement on the metered tape and what bank the measurement begins on (right or left). Depth and discharge will be zero.
- Take the next measurement just off the bank (i.e., closest you can be to the bank while still in the water). Record the distance on the metered tape and measure the depth of the water.
- Determine the depth for velocity measurement readings.
    - If the water depth is < 0.75m, take one measurement at 60% below the water surface.
    - If the water depth is > 0.75 m, take measurements at 20% and 80% of the total depth (80% below the water surface = 20% of total depth, 20% below the water surface = 80% of total depth).
- Adjust the smaller, calibrated rod to the correct setting by lining up the values marked on the rod (0.10 m increments) with the values marked on the handle of the top-setting rod (0.01 m increments). The rod is already calibrated to 60% of the depth from the stream surface. For example if the water depth is 0.5 m on the larger rod, to set the sensor at 60% of the total depth from the surface you need to adjust the thinner graduated rod to also read “5” on the smaller rod, set to the “0” on the rod’s handle. If using 80% and 20% measurements, the thinner graduated rod on the pole can be used to determine adjusted depths by multiplying the calibrated 60% depth by two to obtain the 20% reading and dividing the 60% depth in half to obtain the 80% reading (Table A-7).

Table A-7: Vernier settings for top-setting rod (Source: RISC 2018)

Vernier Setting	Actual Current Meter Position
Observed water depth	0.6 depth from water surface or 0.4 depth up from streambed
Twice observed water depth	0.2 depth from water surface or 0.8 depth up from streambed
Half observed water depth	0.8 depth from water surface or 0.2 depth up from streambed

- Obtain the velocity by recording the observed velocity averaged over 40-70 seconds. The flow sensor should be parallel to the water surface and perpendicular to the measuring tape.
  - Ensure that the person in the stream stands downstream and offset of the current meter so they do not interfere with the measurement.
  - Record notes that will help identify any measurements that may seem uncharacteristic or that may be influenced by an object in the stream channel. For example, negative velocities occur in eddies, which often form behind rocks and/or stream banks. If negative velocities are noticed, record this on the datasheet.
  - Re-measure readings that appear erroneous during the measurement.
- Repeat this procedure, recording the distance on the metered tape, the water depth, and the average velocity reading at each vertical until the watercourse is traversed and the measurement is completed. Record the total discharge if provided by the current meter.
- After completing the measurement, note and record the time and gauge height reading.

A.2.8 Photos

Take photos to provide a visual representation of the measurements and work that have been conducted at the site including:

- View of the hydrometric station facing upstream, downstream, and from each side of the channel
- View of the lake outflow if the station is in a lake
- View of the measurement transect looking upstream, downstream, and from each side of the channel
- General site layout showing site access, nearby infrastructure etc.
- Photos that may help identify any anomalies in the data (e.g., obstructions within the channel)

A.3 Maintenance/Mid-deployment Checks and Data Offload

This section describes a typical workflow for repeat visits to the station once it has been installed. In general, technicians should aim to re-visit the site a minimum of 6 times per year (with one visit at high flow and one visit at low flow).

Step 1 - Review the State of the Hydraulic Control

The first task is to visually examine the condition of the hydraulic control (i.e., the condition downstream from a gauging station that determines the stage-discharge relation).

For hydrometric studies focused on broad streamflow trends, it may be appropriate to ‘rectify’ changes to the control. One example of a control alteration is a log jammed against or downstream of the control resulting in an elevation in stage within the gauge pool. Other examples might include vegetation growth, bank erosion, or the addition or removal of boulders and cobbles.

However, for restoration-focused monitoring—where stations are used to assess changes in stage or flow associated with restoration actions, it is likely inappropriate to interfere with the hydraulic control. Changes may be intentional or unintentional outcomes of restoration, or part of natural background variation. In these cases, control changes should be documented but left undisturbed unless intervention is part of a defined adaptive management plan. This is particularly important for projects involving process-based recovery.

All control alterations or obstructions should be documented and photographed (see Appendix E). If removal is consistent with study objectives, it should only proceed if the obstruction is the sole cause of the alteration, and it can be safely removed.

Technicians should record detailed notes (including the time of obstruction removal), and take photographs to document the removal activity. It is important to wait for stage within the gauge pool to settle prior to starting any further work. If the obstruction cannot be removed, technicians should take photographs and detailed notes recording its likely impact to the stage record and the stage-discharge rating curve.

Step 2 - Physically Inspect Sensors and Benchmarks

Field technicians should check that sensors are fastened securely and no slippage or loosening has occurred. For the purpose of confirming sensor depth, a measurement of the approximate depth of the sensor should be obtained and the time the depth was taken recorded.

Benchmarks should be inspected for the condition of the mark label to ensure it remains clearly identifiable. In addition, each benchmark should be photographed with sufficient background detail to allow for identification during data review/analysis.

Step 3 - Measure Starting Gauge Height

Gauge height can be obtained from either (1) direct visual observation of a staff gauge or (2) levelling surveys from benchmarks (i.e., direct water level survey). This must be conducted before starting any other monitoring activities.

The note-taker should record:

- Exact time of measurement
- Name of the person taking the measurement
- Photographs of the measurement as it is being taken, with emphasis on ensuring readability of measured values (for staff gauge observation).

Step 4 – Perform a Levelling Survey (as necessary)

Levels should be run between all benchmarks and reference gauges once a year at stable sites (ideally after spring thaw), and two times per year at more unstable sites. When running levels to check for the possible movement of a reference gauge, the circuit must be closed and closure error must be within survey tolerance. Any changes should be documented in the station history.

Step 5 – Measurement of Discharge

The field team should follow procedures to obtain discharge measurements (Section A.2.7).

The note-taker should record:

- Time at the start and end of the measurement
- Name of the person completing the measurement
- Measurement-specific information
- Qualitative assessment of the quality of each measurement

- Photographs or videos of the discharge and the discharge measurements

Refer to field forms in Appendix E.

Step 6 – Measure Final Gauge Height

Monitoring activities should conclude with the acquisition of a final gauge height, using the same procedure outlined in Step 3.

Step 7 – Real-time Data and Download

One of the last tasks for any site visit is to view real-time stage readings from the datalogger and download the datalogger. This should be done last as the time-series leading up to the end of the site visit will be captured within the data record, so interpolation during data processing following the site visit is straightforward in situations where the gauge height may have been changing during the visit.

While viewing real-time data and downloading data, the following routine checks should be completed:

- Power supply – In instances where power is supplied to the datalogger internally, a check may be performed when viewing real-time stage immediately prior to, or following the data download. For externally powered dataloggers, a hand-held volt meter can be used to check the power of the main battery. If a solar panel is installed, the volt meter can be used to test the solar output. Note the condition of all power on the field form (Appendix E).
- Memory Capacity – Check the remaining memory within the datalogger. Note this on the field form.
- Desiccant – Some instruments require desiccants to keep the electronic components dry/free from condensation. Note the state of the desiccant on the field form and change it out as necessary (fresh desiccant is often brightly coloured and turns pink to clear as it becomes stale/saturated with water).
- Datalogger clock – compare the real-time clock with the correct standard time. If the clock has deviated a reset might be required. Clock resetting can result in clearing of the datalogger memory so technicians should ensure that all data are downloaded from the datalogger prior to resetting the clock.

Step 8 – Station Maintenance

A physical check of all station components (e.g., electrical wiring, rubber, plastic or metal components, external batteries etc.) should be completed throughout the site visit. Replace components as necessary.

Step 9 – Post retrieval QAQC Procedures for Pressure Transducer Data (Office-based Task)

Quality Assurance and Quality Control (QAQC) is essential to collecting accurate long-term data. The guidance presented below (adapted from USEPA 2014), outlines a standard set of procedures that should be performed to verify the quality of the data and to detect and correct for potential errors.

Record Keeping and Data Storage

- **Set up a robust record keeping and data storage system**. Large amounts of data will accumulate quickly, so it is important to develop and maintain a central database.
- **All field forms should be organized**, accessible and archived in a way that allows for safe, long-term storage.

- **Original raw data files should be retained** for all sites and should be kept separate from files in which data have been manipulated.
- **Carefully document any changes** that are made to the data. The checks can be conducted using a number of different software packages (e.g., Microsoft Excel, Hoboware, Aquarius). Both the original and the "cleaned" data files should be maintained and backed up.

Data Evaluation

1. Save the file that you are manipulating with a **different file name** so that it is not confused with the original raw data file.
2. Format the data so that it is easy to work with.
3. Trim data (as necessary) to remove measurements taken before and after the pressure transducer is correctly deployed. This can be done via a visual inspection of data and by referencing field notes indicating the exact times of deployment and recovery. While reviewing the field notes, also look for comments about situations that could cause the transducer to record questionable readings (e.g., during a mid-deployment check, the transducer was found to be buried in the sand) and flag those data accordingly.
4. Plot all stage measurements and visually check the data. Look for missing data and abnormalities. Watch for:

    a. Missing data

    b. **Values of 0** – this could mean that the pressure transducer was dewatered. Another possibility (with vented transducers) is that moisture got into the cable and caused readings of zero water depth.

    c. **Values flat-lining at 0°C** – the stream pressure transducer is likely encased in ice.

    d. **Negative values** – if unvented transducers are being used, this may indicate that the barometric pressure correction is off. This could occur for a number of reasons:

        i. The land-based transducer is not close enough to the stream pressure transducer to accurately capture barometric pressure.

        ii. If the land-based transducer is housed in PVC pipe that has a solid bottom, condensation and laterally blown rain and snow can penetrate through the drilled holes and collect in the bottom, filling the pipe to a depth sufficient to inundate the ports through which the barometric pressure is compensated. Thereafter, "barometric pressure" is actual barometric pressure plus a small amount of pressure due to this accumulated water.

    e. **Outliers or rapidly fluctuating values** – the stream pressure transducer may have moved (e.g., due to a high flow event or vandalism), or may be encased in ice.

5. Plot stage and temperature data on the same graph. Watch for the following signals in the temperature data:

    a. **Diel fluxes with flat tops** – this indicates that the pressure transducer may have been buried in sediment.

    b. **A close correspondence between water and air temperature** (if air temperature data are available) – this indicates that the pressure transducer may have been out of the water.

6. Graphically compare data across years.
7. Graphically compare with precipitation data from the nearest active weather station, if appropriate.
8. Graphically compare with data from the nearest stream gauge, if appropriate.

Application of Data Corrections

Erratic readings with pressure transducers can occur for a number of reasons:

- They may become dewatered during low flow conditions
- High flow events may bury them in sediment
- High flow events may move them
- They may become fouled from debris, aquatic vegetation, or algae
- Humans and/or wildlife may cause interference
- They may become encased in ice
- If moisture gets into the cable of a vented transducer, it may result in erratic readings or readings of zero water depth
- If the cable of a vented transducer gets kinked or plugged, it can result in the data not being corrected for barometric pressure

Errors should be addressed on a case-by-case basis. In general, there are three possible actions:

1. Leave the data as is
2. Apply the correction factor
3. Remove the data

Various software applications can be used to help with the error screening checks (e.g., Microsoft Excel, Aquatic Informatics' AQUARIUS, including software that is purchased with the transducers). If you are inexperienced at addressing errors with continuous temperature data, consider seeking guidance from someone with more experience and consult references like Wagner et al., (2006) and ECCC (2022). Table A-8 provides a general summary of different types of problems that can occur (e.g., missing data, failed accuracy check) and recommended actions for addressing them. **Corrections should not be made unless the cause(s) of error(s) can be validated or explained in the field notes or by comparison with information from nearby stations.**

Table A-8: General summary of different types of problems that can occur with pressure transducer data and recommended actions for addressing them (Source: USEPA 2014)

Problem	Recommendation
Missing data	Leave blank
Stream pressure transducer was dewatered or buried in sediment for part of the deployment period	Use the plot (and temperature data, if available) to determine the period during which the problem occurred. Exclude these data when calculating summary statistics.
Recorded values are off by a constant, known amount (e.g., due to a calibration error)	Adjust each recorded value by a single, constant value within the correction period.
Drift is large and when and how much the sensor was 'off' cannot be determined (when drift occurs, the difference between staff gauge or depth readings and pressure transducer readings increases over time)	The data should be removed.
Discrepancy between pressure transducer reading and discrete measurement taken during a staff gauge or depth check	General rules: ■ If the errors are smaller than the accuracy quoted by the manufacturer and cannot be easily corrected (e.g., they are not off by a constant amount), leave the data as is, and include the data in summary statistics calculations. ■ If the transducer fails a staff gauge or depth accuracy check, review field notes to see if any signs of disturbance or fouling were noted, and also look for notes about the quality of the gauge measurement (Were flows fluctuating rapidly at the time of the measurement?). Also check whether the same time setting was used for both the transducer and gauge or depth measurements (daylight savings time vs. standard time). Based on this information, use your best judgment to decide which action (leave as is, apply correction, or remove) is most appropriate.
A shift is detected and an elevation survey reveals that the stream pressure transducer has moved	Stage readings can be adjusted by adding or subtracting the difference in elevation. If the exact date of the elevation change is unknown, compare gauge data to transducer data to observe any shifts. If no gauge data for the time period are available, transducer data should be examined for any sudden shifts in stage. Changes in the elevation typically occur during high flows, so closely examine all data during these time periods.
The sensitivity of the pressure transducer changes with stage (e.g., the transducer is less sensitive or accurate at high stages)	Sensitivity drift can be detected by graphing the difference between transducer and staff gauge readings against the gauge height and plotting a linear trend line through it. A strong correlation between the data sets and a positive or negative trend line as stage increases or decreases could indicate a sensitivity shift. Based on this information, use your best judgment to decide which action (leave as is, apply correction, or remove) is most appropriate.

APPENDIX B: TWO-PEG TEST PROCEDURE

The two-peg test is a simple closed circuit level run used to ensure the line of sight through the telescope of a level is horizontal. The procedure, adapted from ECCC (2023), is as follows:

1. Establish two points, A and B, about 60 - 90 m apart on level ground and mark each point with a peg or stake.
2. Set up and level the instrument midway between points A and B.
3. Stand levelling rod on point A and take reading (a)
4. Stand levelling rod on point B and take reading (b)
5. Record the difference in these readings (i.e., b - a). Since the observations are made from a point that is equidistant from each stake, the difference in the reading (b-a) is the correct difference in elevation between the two stakes, regardless of any error in the instrument.
6. Set up and level the instrument as close as possible to point A.
7. Stand levelling rod on point A and take reading (c)
8. Stand levelling rod on point B and take reading (d)
9. If the instrument is in good adjustment, the difference in elevation of the two stakes as observed from stake A will be the same as when observed from midway between them, so that d-c will equal b-a. The correct reading of the rod at stake B (e) is equal to b - a + c. Therefore, the difference between this value and the actual reading at d is the error in the adjustment of the line of sight between the two stakes (I.e., the collimation error). If the collimation error exceeds 0.001 m over a 30 m sighting distance, the instrument should be sent to a calibration facility for proper adjustment as per the manufacturer's instructions.

$$\text{Collimation error} = b - a + c - d$$

Table B-1 provides an example of a two-peg test conducted with an instrument in proper adjustment (Example 1) and with an instrument that is not in proper adjustment (Example 2).

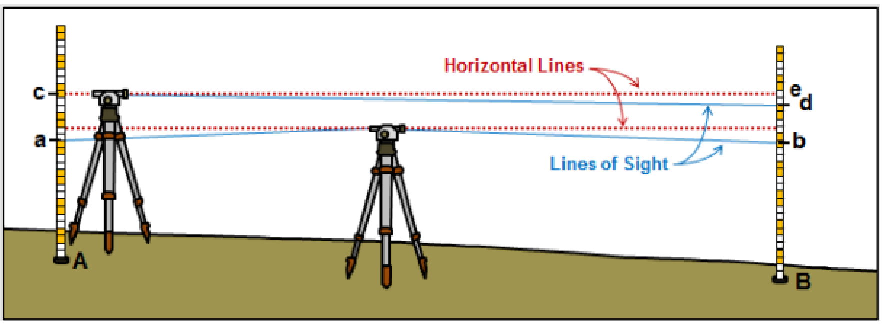


Figure B-1: Illustration of the principles of the two-peg test (Source: ECCC 2023)

Table B-1: Two-Peg Test Examples

Position	Example 1: Level IN adjustment	Example 2: Level NOT IN adjustment
Distance between rods A and B	90 m	90 m
a	1.510 m	1.390 m
b	2.230 m	2.110 m
b – a	0.720 m	0.720 m
c	1.730 m	1.630 m
d	2.450 m	2.310 m
d - c	0.720 m	0.680 m
(Difference b-a) – (Difference d-c)	0.000 m	0.040 m
Collimation error over a 30 m sighting distance	0.000 m	0.013 m
Less than or equal to 0.001 m/30 m	Yes	No

APPENDIX C: PRESSURE TRANSDUCER TYPES

Table C-1: Overview of vented and non-vented pressure transducers (Adapted from: RISC 2018 and Associated Environmental 2017)

Instrument Type	Preferred Installation	Range of Measurement	Resolution and Accuracy	Programming and Datalogging Options	Power Supply Options	Maintenance and Calibration Interval	Advantages	Limitations
Submersible Pressure Sensors – Non-vented	▪ Instream – sensor inside stilling well* (e.g.,pipe, culvert) ▪ Installed in sufficient water depth to measure full range of stage ▪ Fresh water applications	▪ Min 3.5 mm ▪ Max >4.0 m	▪ Resolution = 1 mm ▪ Accuracy = ±0.15% Full Scale ▪ Accuracy based on combined errors from water level and barometric pressure sensors	▪ Self-contained datalogger (instream) with manual communication; or ▪ External communication cables available; or ▪ User able to program sampling rate as well as averaging intervals; or ▪ Software should compensate water level values for changes in barometric pressure from an external source	▪ Internal battery; or ▪ 12V battery converter; or ▪ AC / Solar power options (if part of monitoring design)	▪ Sensor will drift – compared to actual stage reading each site visit ▪ Calibrate as per manufacturer's recommendations (generally every 1–2 years)	▪ Can be used under all streamflow conditions. ▪ Available in both cabled and non-cabled versions. ▪ Sensors are totally sealed against water ingress. ▪ Stand-alone systems are very compact allowing them to be deployed inconspicuously. ▪ Real-time communication with sensor and data download with cable option. Thus, sensor remains in place. Loggers can now be programmed by Bluetooth, laptop, and miscellaneous connectors	▪ Requires external compensation for changes in barometric pressure. Most software requires barometric correction from same make or specific models or loggers. ▪ Pressure diaphragm can be damaged if left to freeze in shallow waters during winter. ▪ Decrease in overall system accuracy due to the use of two sensors (water level and barometric pressure). ▪ Stand-alone systems typically need to be extracted from the water to recover data, which may be problematic depending upon site and

							using tablets or smartphones. ■ Stand-alone systems (i.e., self-contained datalogger) are relatively inexpensive.	streamflow conditions. ■ Internal battery cannot be replaced. ■ For self-contained datalogger sensor options, risk of records loss if sensor is lost or damaged due to high water, vandalism, etc. ■ Limited sensor datalogging options (i.e., non-flexible recording options).
Submersible Pressure Sensors – Vented	■ Instream – sensor inside stilling well* (e.g.,pipe, culvert) ■ Installed in sufficient water depth to measure full range of stage ■ Fresh water applications	■ Min 3.5 mm ■ Max >4.0 m	■ Resolution = 1 mm; ■ Accuracy = ± 0.1% Full Scale	■ Self-contained datalogger (instream) with external communication; or ■ Output signal to an external datalogger (in enclosure) via hardwired connection; or ■ Able to set water level, sample rate, averaging period through software	■ Internal battery; or ■ 12V battery converter; or ■ AC / Solar power options (if part of monitoring design)	■ Sensor will drift – compare to actual stage reading each site visit; ■ Calibrate as per manufacturer's recommendation (generally every 1–2 years)	■ Can be used under all streamflow conditions. ■ Real-time communication with sensor and data download. Thus, sensor remains in place. ■ Greater system accuracy (in comparison to non-vented sensors) as barometrically corrected. ■ Real-time telemetry options available with cellular modems, radio, and satellite transmitters (if part of	■ Presence of vent line requires more work to install, as vent line needs to be protected against insect inhabitation, debris, etc., which may block the line and affect barometric compensation. ■ Pressure diaphragm can be damaged if left to freeze in shallow waters during winter. ■ May need backup barometric data in case of line failure. ■ Periodic changing of desiccant is required to keep vent tube free of moisture.

							monitoring design).	■ Vent line poses snag risk for debris in stream. ■ Possibility of moisture entering internal electronics and damaging them if the cable integrity is compromised. ■ Internal battery option not ideal for extended deployment. Battery replacement under this option is difficult to complete in the field. ■ For self-contained datalogger sensor options, risk of records loss if sensor is lost or damaged due to high water, vandalism, etc.

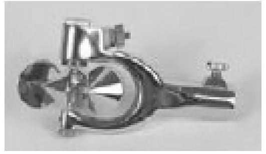
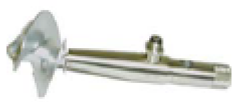
**Stilling wells are vertical, pipe-shaped enclosures placed in or near the streambank. They are used in hydrometric monitoring operations to dampen water level fluctuations and protect sensor components. 3” diameter aluminum or ABS pipe can be used as stilling wells. Note that all stilling wells should be open at the top, or have screen-covered vent holes to allow humidity to escape and to ensure pressure within the well is at barometric pressure.*

APPENDIX D: CURRENT METER TYPES

- Mechanical current meters
    - Measure velocity by translating linear motion into angular motion. The rate of rotation of the rotor or propeller is used to determine the velocity of the water at the point where the current meter is set.
    - Two common types: Vertical-axis meter and the horizontal-axis meter.
    - Before the current meter is used, the relationship between the rate of rotation and the velocity of water is established in a towing tank.
- Electromagnetic current meters
    - Measure velocity using Faraday's principle (electromagnetic induction), which states that a moving conductor (i.e., water) in a magnetic field will generate a voltage proportional to the speed of the conductor.
    - Electrodes in the probe detect a voltage induced by the flowing water and convert the voltage into a velocity reading.
    - These meters have no moving parts and are therefore not subject to many of the operational and maintenance problems associated with mechanical current meters.
    - The electromagnetic flow sensors are designed for the measurement of a wide variety of flow magnitudes, including very low flow velocities and for site conditions where the conventional current meters can no longer be used (i.e., areas with large numbers of aquatic plants, contaminated water, shallow water, and low-velocity water).
- Acoustic doppler velocity (ADV) meters
    - ADV meters determine water velocity by measuring the change in acoustic frequency (or Doppler shift) from reflections from moving particles or scatters (such as suspended sediment) in the flow, which are assumed to be moving at the same velocity as the water.
    - In comparison to mechanical cup meters, the ADV meter offers advantages such as a wider velocity range, and measurement in more shallow water.
    - Acoustic meters can be affected by both high and very low levels of suspended solids or too much entrained air; therefore, site selection is important.
    - The Water Survey of Canada has officially adopted the use of ADVs (specifically Sontek's FlowTracker ADV) to support their national hydrometric monitoring program and have established standardized field procedures (i.e., "Measuring Discharge with FlowTracker Acoustic Doppler Velocimeters", Environment Canada 2015) for conducting a discharge measurement using wading techniques.

Table D-1: Comparison of Discharge Measurement Equipment and Methods (Adapted from: RISC 2018 and Associated Environmental 2017)

Instrument Type	Technology	Deployment Method	Range of Measurement	Accuracy	Maintenance/Calibration	Advantages	Limitations
Price Type AA 	■ Mechanical ■ Manual counting or digital readout options ■ Digital readout battery powered	■ Mid/Mean-section method using top-setting rod, ice rod, or cable and weight assembly ■ Field applications: Wading, bridge, boat, cableway, ice	■ •Velocity: 0.06 – 7.6 m/s ■ Water Depth: >0.15 m	■ Accuracy = ±2.6% in velocities from 0.229 m/s to 2.438 m/s	■ Meters should be calibrated every 2–3 years (depending on level of use), unless meter has been damaged. ■ A spin test should be conducted prior to using to ensure that the bucket assembly rotates easily.	■ Robust and simple to use. ■ Does not have to be oriented parallel to streamflow to read accurately. ■ Repeatable readings. ■ Recognized as standard equipment by the Water Survey of Canada (WSC). ■ Can be calibrated to National Calibration Services Standards.	■ Requires regular maintenance to ensure minimal friction in operation. ■ Possible human error with older “click” systems. ■ Does not give flow angles, requires knowledgeable operator. ■ May not be suitable for shallow or narrow channels. ■ Not ideal for very low flows. ■ Off-axis (oblique) streamflows must be manually corrected. ■ Debris/vegetation within the water column can affect meter readings.

Pygmy Meter 	■ Mechanical ■ Manual counting or digital readout options. ■ Digital readout battery powered	■ Mid//Mean-section method using top-setting rod. ■ Field applications: wading	■ Velocity: 0.02 to 0.9 m/s ■ Water Depth: 0.076 m up to height of rod	■ Accuracy = ±4% in velocities from 0.229 m/s to 0.914 m/s	■ Meters should be calibrated every 2–3 years (depending on level of use), unless meter has been damaged. ■ A spin test should be conducted prior to using to ensure that the bucket assembly rotates easily.	■ Robust and simple to use. ■ Intended for use in small channels. ■ Does not have to be oriented parallel to streamflow to read accurately. ■ Repeatable readings. ■ Recognized as standard equipment by the WSC. ■ Can be calibrated to National Calibration Services Standards.	■ Off-axis (oblique) streamflows must be manually corrected. ■ Variability in accuracy depending upon velocity. ■ Debris/vegetation within the water column can affect meter readings.
Horizontal Axis Meter – Component Propeller 	■ Mechanical ■ Digital (counting) readout ■ Digital readout battery powered	■ Mid/Mean-section method using top-setting rod. ■ Field applications: wading	■ Velocity: 0.02 – 0.9 m/s ■ Water Depth: 0.03 m (depending on propeller type) up to height of rod	■ Accuracy = ±2% (when using a component propeller and provided with a rating from the National Calibration Service)	■ Meters should be calibrated every 2–3 years (depending on level of use), unless meter has been damaged. ■ A spin test should be conducted prior to using to ensure that the propeller assembly rotates easily.	■ Propeller disturbs streamflow less than the Price Type meter. ■ Propellers are less prone to becoming entangled with debris. ■ Lower starting threshold	■ A propeller's response to increases in velocity is non-linear, so a multi-point calibration should be done for entire measurement range. ■ Higher maintenance when compared to

						compared to vertical axis meters. ■ Some propellers can be specifically chosen to suit shallow waters, low streamflows, or turbulent conditions. ■ Component propellers are capable of compensating for off-axis (oblique) streamflows ■ Can be calibrated to National Calibration Services Standards.	vertical-axis meter as the meter should be serviced at the end of each discharge measurement.
Electromagnetic Flow Meters 	■ Electronic ■ Digital measurement and datalogging	■ Mid/Mean-section method using top-setting rod. ■ Field applications: wading	■ Velocity: 0.01 – 6 m/s ■ Water Depth: 0.032 m up to height of rod	■ Accuracy = 2% to 4% (based on velocity measurement)	■ Considered permanently calibrated by manufacturer. ■ Run self-check diagnostics prior to field use to ensure proper instrumentation operation.	■ No moving parts to wear out. ■ Calibration will not change over time. ■ Able to measure low flows. ■ Direct velocity and	■ Sensor performance and accuracy can be affected in low electrical conductivity waters (<200 μS/cm). ■ Proximity to ferrous material boundaries(e.g., culverts, metal grates) can

						discharge readout. ■ Works well in high sediment conditions, turbulent waters, and areas subject to aquatic plant growth.	affect the measurement accuracy. ■ Expensive.
Acoustic Doppler Velocimeters 	■ Electronic ■ Digital measurement and data logging	■ Mid/Mean-section method using top-setting rod. ■ Field applications: wading or deployment from ice	■ Velocity: 0.001 – 4 m/s (Point velocity) ■ Water Depth: ○ 0.02 m (Point velocity) up to height of rod ○ 0.05 m (Pulse Coherent) up to height of rod	■ Accuracy = ±1% (of measured velocity)	■ Considered permanently calibrated by manufacturer. ■ Run self-check diagnostics prior to field use to ensure proper instrumentation operation.	■ Significant advantage in software for data collection, real-time error warnings and minimizing data transfer errors. ■ No moving parts to wear out. ■ Calibration will not change over time. ■ Able to measure low flows – point velocity style has better accuracy	■ Clear water conditions are a challenge for Doppler instruments because they require particulate matter in the water to reflect sound (point velocity type less so than the pulse coherent type). ■ Aquatic vegetation can interfere with measurement. ■ May not be suitable at low flows where substrate is coarse; requires clear line of site of about 10 cm from sensor

						and resolution. ■ Direct velocity and discharge readout. ■ Works well in high sediment conditions. ■ Modern digital units have many advantages in data post-processing	unit to measurement volume. ■ Does not perform well in turbulent conditions. ■ Expensive.

APPENDIX E: HYDROMETRIC MONITORING FIELD FORMS

Project Name:		Station Name:	
Stream Location:		Date (YYYYMMDD):	
Arrival Time (24hh:mm)		Departure Time (24hh:mm):	
Weather (e.g., recent rain or current weather):			
Personnel:			
Water Level Gauge			
DL Model:		PT Serial #:	
Gain:		Offset:	
Status:		Battery:	
Readings Taken:		Readings Remaining:	
Manual Gauge:		Memory Cleared (Y/N):	
Station Coordinates (UTM)			
Zone:			
Easting:			
Northing:			
Elevation:			
Photographs			
	Photo Number(s)	Notes	
Upstream:			
Downstream:			
Left Bank:			
Right Bank:			
Gauging Section:			
Control (s):			
Staff Gauge:			
Pressure Transducer:			
Benchmarks:			

Notes - Maintenance, station stability, changes to control features and channel reach

Site Sketch - Including channel conditions and control features, site access, major landmarks, equipment, benchmarks etc.

Levelling Survey Form

Project Name:		Station Name:	
Date (YYYYMMDD):		Time (24hh:mm):	
Real Time Stage (3 readings):		Staff Gauge (min, max, avg):	
Surveyed by:		Instr. Leveled	

Station	BS (+)	Hi. Inst. (m)	FS (-)	Elevation (m)	Notes

Elevation Means

Benchmark #	Established Elevation (m)	Mean Elevation [this date] (m)	Difference (m)

Notes:

Discharge Measurement Form (Complete both sides)

Project Name:		Station Name:	
Date (YYYYMMDD):		Metered by:	
Weather (e.g., recent rain or current weather):			
Meter Type:		Avg. Interval(s) per Reading:	
Serial #:		Meter Calibrated (Y/N, Date):	
Channel condition or other condition affecting control or discharge measurements (e.g., variable backwater, turbulence, vegetation etc.):			

Station Location		
Zone:		Location of Metered Section (description):
Easting:		
Northing:		
Elevation:		

Stage Readings				
	Real Time Stage (from datalogger reading; m)	Staff Gauge (m)	Time (24hh:mm)	Notes
Begin				
End				

Discharge Measurement Form (Complete both sides)

Station #	Distance (m)	Depth(m)	Velocity (m/s)			Notes
			60%	20%	80%	
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						
21						
22						
23						
24						
25						
26						
27						
28						
29						
30						
Total Q						

Notes: